COVER SHEET FOR PROPOSAL TO THE NATIONAL SCIENCE FOUNDATION

PROGRAM ANNOUNCEMENT/SOLICITATION NO./DUE DATE	☐ Special Exception to Deadline Date Policy	FOR NSF USE ONLY
NSF 19-528 11/15/19		NSF PROPOSAL NUMBER
FOR CONSIDERATION BY NSF ORGANIZATION UNIT(S) (Indicate the most specific unit known, i.e. program, division, etc.) BCS - DYN COUPLED NATURAL-HUMAN		

DATE RECEIVED	NUMBER OF COPIES	DIVISION ASSIGNED	FUND CODE	DUNS# (Data Universal Numbering System)	FILE LOCATION
				041968306	

EMPLOYER IDENTIFICATION NUMBER (EIN) OR TAXPAYER IDENTIFICATION NUMBER (TIN)	SHOW PREVIOUS AWARD NO. IF THIS IS ☐ A RENEWAL ☐ AN ACCOMPLISHMENT-BASED RENEWAL	IS THIS PROPOSAL BEING SUBMITTED TO ANOTHER FEDERAL AGENCY? YES ☐ NO ☒ IF YES, LIST ACRONYM(S)
135562308		

NAME OF ORGANIZATION TO WHICH AWARD SHOULD BE MADE	ADDRESS OF AWARDEE ORGANIZATION, INCLUDING 9 DIGIT ZIP CODE
New York University	New York University 70 Washington Square S New York,NY.100121019
AWARDEE ORGANIZATION CODE (IF KNOWN) 0027854000	
NAME OF PRIMARY PLACE OF PERF	ADDRESS OF PRIMARY PLACE OF PERF, INCLUDING 9 DIGIT ZIP CODE
New York University	New York University 44 West 4th Street New York ,NY ,100121276 ,US.

IS AWARDEE ORGANIZATION (Check All That Apply) ☐ SMALL BUSINESS ☐ MINORITY BUSINESS ☐ FOR-PROFIT ORGANIZATION ☐ WOMAN-OWNED BUSINESS ☐ IF THIS IS A PRELIMINARY PROPOSAL THEN CHECK HERE

TITLE OF PROPOSED PROJECT **CNH2-L:Transforming Tropical Commodity Supply Chain Partnerships to Foster Carbon Stocks and Sustainable Livelihoods**

REQUESTED AMOUNT	PROPOSED DURATION (1-60 MONTHS)	REQUESTED STARTING DATE	SHOW RELATED PRELIMINARY PROPOSAL NO. IF APPLICABLE
$ 1,545,328	36 months	09/01/20	

THIS PROPOSAL INCLUDES ANY OF THE ITEMS LISTED BELOW

☐ BEGINNING INVESTIGATOR
☐ DISCLOSURE OF LOBBYING ACTIVITIES
☐ PROPRIETARY & PRIVILEGED INFORMATION
☐ HISTORIC PLACES
☐ VERTEBRATE ANIMALS IACUC App. Date _____
PHS Animal Welfare Assurance Number _____
☒ TYPE OF PROPOSAL Research

☒ HUMAN SUBJECTS Human Subjects Assurance Number FWA00006386
Exemption Subsection _____ or IRB App. Date 11/13/19
☐ FUNDING OF INT'L BRANCH CAMPUS OF U.S IHE ☐ FUNDING OF FOREIGN ORG
☒ INTERNATIONAL ACTIVITIES: COUNTRY/COUNTRIES INVOLVED
GH
☒ COLLABORATIVE STATUS
Not a collaborative proposal

PI/PD DEPARTMENT	PI/PD POSTAL ADDRESS
PI/PD FAX NUMBER	44 West 4th St. New York,NY 10012 United States

NAMES (TYPED)	High Degree	Yr of Degree	Telephone Number	Email Address
PI/PD NAME Tracy Van Holt	PhD	2009	212-998-0226	tvanholt@stern.nyu.edu
CO-PI/PD Craig Carter	PhD	1996	480-965-6044	craig.carter@asu.edu
CO-PI/PD Matthew N Hayek	PhD	2017	202-746-5522	matthew.hayek@nyu.edu
CO-PI/PD Jeffrey C Johnson	PhD	1981	252-258-4869	johnsonje@ufl.edu
CO-PI/PD				

CNH2-L:Transforming Tropical Commodity Supply Chain Partnerships to Foster Carbon Stocks and Sustainable Livelihoods

PROJECT SUMMARY

Overview:

Food companies around the world are exploring how best to create strategic partnerships that provide farmers with more equitable pricing and contracting standards that in turn help them implement practices reducing environmental impacts. The proposed research investigates the Coupled Natural-Human Systems (CNHS) of smallholder farmers in tropical cocoa agroforestry systems in Ghana and their strategic partnerships with global commodity firms. The objectives of this project are to (1) understand why farmers decide to engage or not in strategic sourcing partnerships; (2) assess whether strategic partnerships can translate into landscape-wide climate change mitigation and sustainable livelihoods; (3) evaluate in financial terms the sustainability benefits generated by such partnerships for the supply chain. We will rely on a district-forest field survey comparative design, comparing areas with high/low forest cover and high/low levels of strategic partnerships, to isolate the respective role of environmental conditions and social networks on environmental outcomes and farmer participation. This study will combine land-use change, cultural consensus and network analysis to theorize about the individual and contextual characteristics leading farmers to have certain participation profiles (logical, peripheral, and follower partners) in strategic partnerships. We will also collect data about farmer's agricultural and management practices to examine the carbon opportunity costs of different types of strategic partnerships and cocoa production systems. Through our modelling of the Ghanaian-wide cocoa supply chain network, a landscape-level analysis and socio-economic demographics data, we will provide carbon stocks estimate for the entire cocoa supply chain in Ghana. We will also estimate whether sustainability benefits depend upon single of multiple strategic partnerships within a given district. Finally, our study will evaluate and quantify the financial benefits of strategic partnerships for both commodity firms and farmers by relying on the NYU's Center for Sustainable Business (CSB) Return on Sustainability Investment (ROSI).

Intellectual Merit:

The proposed research will fill an important gap by linking the content of strategic partnerships to on-the-ground environmental outcomes, such as reforestation and sequestered carbon. The combination of different methods (network, cultural consensus, and land-use analyses) will help demonstrate the respective roles of local environmental conditions and community social networks in determining environmental outcomes and farmer participation. By modelling and monetizing the carbon opportunity costs and other sustainability benefits of different types of strategic partnerships and cocoa production systems for the first time, it will demonstrate the existence of win-win scenarios in which agricultural systems can sustain livelihoods and help mitigate climate change. The national-scale analysis linking supply chain characteristics to estimates of stored carbon will advance the understanding of climate change impacts from agroforestry systems. This research illustrates NSF's Growing Convergence Research idea, according to which the combination of disciplinary theories (forest transition, network, institutional theories) and methods can unlock key insights into the grand challenges of today.

Broader Impacts:

This research project will inform policy and practice on how to structure sustainable supply chains and strategic sourcing partnerships for landscape- and farmer-level environmental and societal benefits. Driving this focus is the concern that if current problematic socio-economic, environmental and agronomic conditions are not addressed, companies will no longer be able to source cocoa from West Africa and farmers will no longer be able to make a living and support their families. This project will offer critical insights to (global and U.S.) companies purchasing commodities, NGOs in charge of sustainability certification, and policy makers in charge of land-use policy, by identifying the key levers and impacts of strategic partnerships as well as monetary estimates of their benefits to help drive more investment. These will include the advisors to the project Mars (private US company) and Rainforest Alliance (US NGO). This project will further interdisciplinary research by offering postdoctoral training based on merging disciplines (supply chain management, geography, ecology, and business) and research methods (network analysis, remote sensing, and cultural consensus analysis). In addition, the project will engage a broad diversity of scholars engaged in CNHS and NYU students, leveraging the NYU Accra satellite office and Ghana Statistical Services, the project partner.

CNH2 components: Conventional, transactional commodity sourcing from small farmers around the world is a major driver of poverty, deforestation, and carbon emissions. In response, food companies are creating strategic partnerships providing farmers with more equitable pricing and contracting standards that should help them implement more sustainable practices that reduce environmental impact. The rise in these strategic partnerships is unmistakable, yet major transformations in our food system remain to be seen at scale. We argue that this is the result of solutions designed without understanding the coupled human-natural system (CHNS) linking supply-chain sourcing strategies of global commodity firms with the agroecosystems of commodity production. Much of the world's commodity markets in tropical countries have similar market pressures and competing land cover/uses. We focus on the small-landholder cacao (*Theobroma cacao*) sector in Ghana as it combines one of the highest rates of deforestation and certification (i.e. strategic partnerships) of any commodity sector.

The **natural component** of our research is the cocoa agroecosystems. Like many agricultural tree crops, tree cover affects soil quality, biodiversity, carbon sequestration, and climate-change adaptation. Whether a farm is located in a **preserved or degraded ecosystem** (see Figure 1, component 1) can determine the quality of the fruit, the resistance of cocoa trees to climate-related weather shocks, and the type of non-cocoa trees providing shade, which influences the productivity and carbon footprint of **cocoa production** systems (component 5). The **human component** of our CNHS is represented by the decisions of supply chain actors, including global corporate buyers, intermediaries such as cooperatives, and farmers. Farmer's decisions on **cocoa production** (5) includes whether they sell their product to the **strategic partnership** or **market-based market** (6 &7). Those companies and farmers interested in strategic partnerships perceive some benefits (10), but struggle with how to monetize these benefits. Once the benefits are monetized, and the CNHS is understood, then investments can be tailored to foster a positive cycle (see the top half of Figure 1). To sell in the strategic partnership market, buyers offer a price premium that can influence decisions.

Our main human-natural system coupling of interest is how farmers' participation in strategic partnerships influence their adoption of **sustainable agricultural practices** (9) which ultimately influences environmental quality. These practices should reinforce a **preserved ecosystem** (1). The practices of those famers in the market-based system may lead to a **degraded ecosystem** (1) because of **unsustainable agricultural practices,** or switching activities to illegal gold mining (8). Farmers' decisions about their **cocoa production** (5), depends on **institutional** factors (2) that include whether farmers view the required sustainable agricultural practices as legitimate, and whether they trust their supply chain partners. **Social-ecological** factors (3) such as personal networks, and **economic** (4) factors such as farm size and land tenure also influence decisions. The main **natural-human system couplings under study is how environmental conditions may influence the type of strategic partnerships entered in by farmers and the type of (non)sustainable practices they adopt.** In some cases, the **degraded ecosystem** (1) makes it impossible for farmers to show improvement in sustainable agricultural practices and such perceptions may be reinforced by their social network. In this case, farmers either avoid strategic partnerships in the first place, or they don't comply with the sustainability requests. We will isolate these effects so strategic partnerships can be designed and tailored for given CHNS conditions.

Project description

Deforestation and perpetual poverty persist in our global food system where commodities are bought at the lowest price for the business model. In this model dominated by global firms, who produces food and how it is produced remains largely invisible to corporate buyers and consumers located in distant places (Gardner et al. 2019; Van Holt and Weisman 2016; Yu, Feng, and Hubacek 2013). Yet, the ever-increasing demand for commodities create challenging economic and institutional conditions that lead farmers to adopt short-term, profit-oriented practices that lead to environmental degradation and social exploitation (Lambin et al. 2018). Global commodity firms under increased scrutiny have started reorganizing their supply chains to find incentives for sustainability practices. They have joined forces with NGOs and governments to create strategic partnerships in which they commit to pay farmers a more equitable price as well as provide a set of other capacity-enhancing investments targeted at improving environmental and ethical standards (Lambin et al. 2018). Despite the rise in strategic sourcing, and large investments by companies and governments alike, such partnership have failed to engage a large number of farmers and have had limited environmental quality and social improvements. Worse still, many companies have not yet seen the value of creating them.

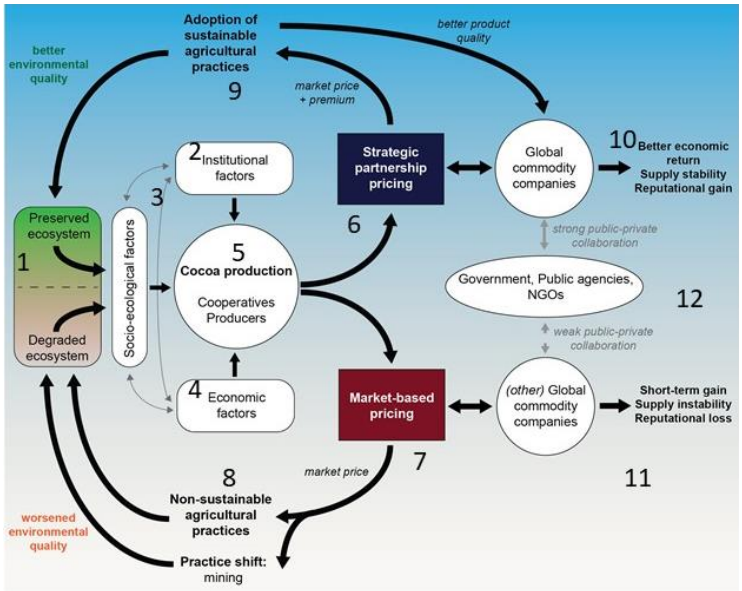


Figure 1: *Coupled Natural-Human System of Cocoa Production in Ghana*

Past research has brought limited insights about how certification schemes result in real environmental benefits, since they essentially focus on measuring on-farm practices or farmer perceptions (Fenger et al. 2017) rather than outcomes (Yoder et al. 2019). The focus of the strategic partnership literature on sustainability certification schemes and their purported productivity and income benefits (Bray and Neilson 2017) to farmers has equally obscured why farmer join such partnerships in the first place and how this decision may be influenced also by environmental, institutional, socio-ecological factors. Social networks may play both a role in farmer participation and practice adoption (Munasib and Jordan 2011; Wossen et al. 2013). The focus on on-farm benefits has also left unexplored how companies creating strategic partnerships perceive and monetize their sustainability benefits (Atz et al. 2019) and how this influence participation (Winters et al. 2015). Finally, the landscape-wide environmental and climate change benefits of strategic partnerships and associated agroforestry commodity production systems they support is one of the key missing information for measuring the ecosystem services and climate change implications of global forest transitions through agroforestry systems (Garrity 2012; Kuyah et al. 2019; Lambin et al. 2018; Tscharntke et al. 2015).

Research goal and questions: Our project contributes to developing a science of strategic partnerships by modeling the institutional, economic, social, and environmental processes and feedback loops that link the agroecosystem of commodity production to global commodity firms into a CNHS. To address the challenges of understanding the complex socio-ecological feedbacks, we investigate the following questions:

1. Why do some farmers engage in strategic sourcing efforts, but not others?
2. Can strategic sourcing efforts increase climate resilience and farm productivity at scale?
3. What are the financial returns of increased climate resilience and productivity for the supply chain?

The project focuses on the smallholder cocoa sector in Ghana. In Ghana the rate of forest loss jumped 60% from 2017-2018—the highest rate increase in all of the tropics (Weisse and Goldman 2019). This is, in part, because cocoa is not perceived as a viable livelihood, despite cocoa accounting for 3.4% of the GDP in Ghana (Läderach et al. 2013). At the same time, Ghana represents the forefront of strategic sourcing partnerships with 50% of cocoa production under some kind of strategic partnership and limited traceability. To understand how Ghana could reach 100%, we adopt a district/forest comparative field survey design. Districts of cocoa production with low and high ecosystem services (native forest cover) and with low and high occurrence of strategic partnerships are compared to determine the relative influences of the natural and human components of the CNHS under study on farmer engagement.

First, this design helps us examining the natural-human coupling of our CNHS by separating the respective role of environmental conditions and community social network characteristics on farmer participation to strategic partnerships. Second, it allows us to calculate the level of carbon stored on-farm depending on partnership and farmer characteristics (including their adopted sustainable practices) and scale up this model at the country-level to explore the landscape-wide implications of cocoa farming for climate change. Finally, we monetize the sustainability benefits of such partnership to reveal hidden drivers of partnership expansion.

Literature Review

Agroforestry, commodity-driven tropical deforestation, and carbon stocks conservation

Just a few agricultural commodities now dominate the global food supply (Gardner et al. 2019; Yu, Feng, and Hubacek 2013) of an ever-increasing world population. Supporting the livelihoods of millions of farmers across the globe, especially in the global South, the soaring expansion of tree-farmed commodities has been heavily associated with deforestation and native forest degradation (Tracy Van Holt et al. 2016; Rudel et al. 2005; Meyfroidt, Rudel, and Lambin 2010; Larsen et al. 2018; Rudel et al. 2019; Sloan et al. 2019).

Humid tropical forests are a critical stock of terrestrial carbon, yet the conversion of native vegetation cover by other types of agricultural uses ends up releasing vast amounts of carbon into the atmosphere (Baccini et al. 2017). Yet the forest conversion implications for carbon stocks depend upon the type of resulting agricultural land-use (Williams et al. 2018). ***In that regard, we know poorly the implications of the conversion of native forest to agroforestry in terms of carbon stocks*** (Garrity 2012; Ordonez et al. 2014). Previous scholarship has discussed the 'carbon opportunity cost' of intensification by comparing carbon in native forests to that of full-sun agricultural production (Searchinger et al. 2018; West et al. 2010). Such costs are likely to vary sensibly from native forests since tree-farms provide different ecosystem services than native forests (Putz and Redford 2010; T. Van Holt and Putz 2017), which has implications for the activities of those dependent on the ecosystem services of native forests (Van Holt 2012; Van Holt et al. 2017). ***These costs must be quantified since they may represent a win-win scenario between agricultural production and climate change for certain crops*** (Williams et al. 2018). In addition, the dynamics of

agroforestry system carbon stocks over time are still poorly understood. Our modeling of carbon dynamics, informed by farmer surveys of farm history and management practices and a literature survey of ecosystem carbon, will address this critical knowledge gap.

Strategic partnerships for sustainability and challenges to sustainable practices adoption

Food companies around the world are creating strategic partnerships providing farmers with more equitable pricing and contracting standards that should help them implement more sustainable practices that reduce environmental impacts, and sequester carbon (Lambin et al. 2018). Yet, despite hundreds of sustainability commitments by companies, few studies evaluate the real social and environmental outcomes of adopted practices. This is partly due to transparency and traceability issues, limited or selective adoption by companies, and deforestation leakage (Gardner et al. 2019; Yu, Feng, and Hubacek 2013; Lambin et al. 2018). ***Past research on strategic partnerships has essentially focused on local level sustainability certification schemes,*** where companies pay a price premium to producers and provide them with training and other kinds of material support in exchange of their adoption of agricultural practices and their commitment to respect environmental and social standards. The results of such schemes have been mixed (Tayleur et al. 2018), and their landscape-wide benefits remain unclear (Tscharntke et al. 2015).

There is in fact considerable uncertainty about how farmers respond to incentives by schemes encouraging (often financially) the adoption of sustainable practices (Prokopy et al. 2008; Baumgart-Getz, Prokopy, and Floress 2012) ***and whether such practices translate into outcomes is rarely measured*** (Yoder et al. 2019). Low levels of practice adoption in Western countries (Carlisle 2016; Kuhfuss et al. 2016; Ribaudo 2015) have led researchers to analyze social-psychological factors such as attitudes, perceptions, specifically on trust and legitimacy, as well as social networks motivate farmers to adopt sustainable practices (Burton 2004; Mills et al. 2017; Spash et al. 2009). In the context of agroforestry farming (coffee, cocoa, etc.) in Western Africa, for instance, price mechanisms too cannot fully explain whether farmer practices will change or not (Akoyi and Maertens 2018) while strong pre-existing institutions within the producer community increase the likelihood of benefits (Bray and Neilson 2017). ***Thus, more research about how farmers' traditional knowledge, perceptions of ecosystem degradation, and interpersonal relationships affect practice change is needed.*** Traditional knowledge may interfere with how experts (i.e. local farmers) perceive resources, thus disregarding requests to change (T. V. Holt et al. 2016; T. V. Holt, Townsend, and Cronkleton 2010; Boster and Johnson 1989). On another level, landscape-level studies have shown that ecosystem services, such as native forest cover play a role in decision making (Van Holt 2012). Social networks matter as well as the type of relations actors have with others determine the legitimacy of requests to change (Munasib and Jordan 2011; Crona et al. 2010). Research has shown that how actors perceive and value relations within the supply chain differs depending on location and role in the commodity chain (Ashby 2016; Zimmerer, Lambin, and Vanek 2018; Youn et al. 2013; Smith 2008; Ki and Rowlinson 2011). It is thus likely that incentives will also vary across social networks ***but to date little empirical work linking different relationship types with incentives to behave sustainably has been done***.

Study Area

With 883 million tons of cocoa produced in 2017, the West African country of Ghana is the second largest producer of cocoa beans in the world. Cocoa cultivation is limited to the southern part of the country where fertile soils are, and about 800,000 to 1,000,000 farmers cultivate the bean (Fenger et al. 2017; Streck et al. 2017). Smallholder plots of two hectares or less with poor or no access to inputs dominate the landscape of cocoa production (Gockowski et al. 2013). The sector is however a major driver of Ghana's economy, supporting the livelihoods of an estimated 6.3 Ghanaians and represent an important share of its export revenue (Glin, Oosterveer, and Mol 2015). **Producers are generally organized into cooperatives, and there may be five to ten of them within a district** (Figure 2). The sector is heavily regulated by the state, which set prices to producers and sets the margins of within-country cocoa purchasing companies. Beyond

minor price bonuses offered by cocoa purchasing firms, farmers can benefit from price premiums if they are engaged into strategic partnerships such as sustainability certification (Fenger et al. 2017; Gockowski et al. 2013). The expansion of cocoa has been linked to extensive deforestation and has shifted of geographical area over time (Gockowski et al. 2013). Droughts in the 1982-1983 periods led to widespread fires in cocoa planting regions of eastern Ghana and triggered mass migration toward the southwestern part of the country where most of the cocoa production takes place today (Ruf, Schroth, and Doffangui 2015). Old extensive shaded production systems have been replaced by intensified full-sun or light-shaded systems (representing an estimated 70% of cocoa production) that contributes to meet the high cocoa production volume target of 1 million tons set by the government (Clough, Faust, and Tscharntke 2009; Gockowski et al. 2013). Yet, the overall declining – or at best stagnant – productivity of most plantations threatens the livelihoods of producers and creates persistent deforestation risks (Streck et al. 2017). Over the past decades, the expansion of farms in Ghana (not only cocoa farms) has contributed to a forest loss of 2% per year with a market over the 2000 to 2011 period (Fountain and Hütz-Adams 2018).

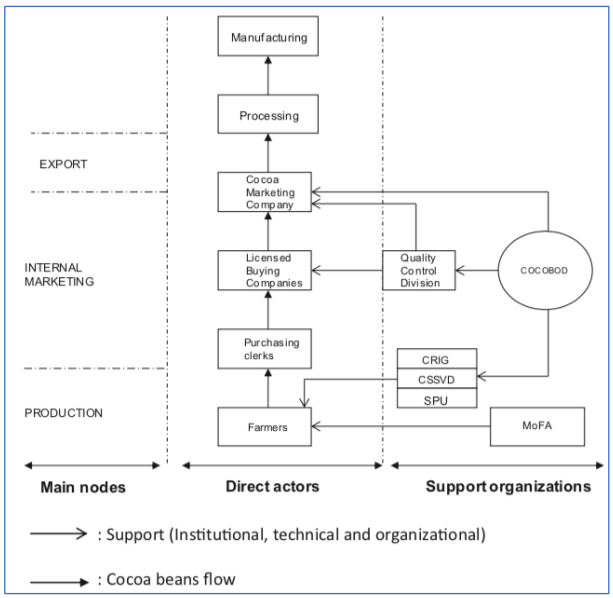


Figure 2 Schematic Cocoa Supply Chain in Ghana (Glin et al. 2015)

In recent years, strategic partnerships have emerged to address the detrimental environmental and social impacts of cocoa production (e.g. child labor) in Ghana. Many companies operating in the US such as Mars, Cadbury, Mendelez and others have funded initiatives encouraging better agricultural practices and/or the certification of these practices under schemes such as Rainforest Alliance, UTZ, or FairTrade (Glin et al. 2015). ***Today, over 50% of Ghana's cacao is under certain kind of certification, but past studies have diverged over the resulting benefits of these initiatives***. Past studies of sustainability certification of cocoa has looked at how certification impacts (1) productivity (Gockowski et al. 2013) and income gains; (2) food security by analyzing the nutrition of farmers (Chiputwa and Qaim 2016); (3) gender equality; (4) perceptions of farmers about economic and environmental change on their farms (Fenger et al. 2017). We still do not know **the factors that have led to the participation of farmers and cooperatives in strategic sourcing** and **whether this has been followed with clearly measurable environmental outcomes**. For instance, studies have fallen short of **measuring concrete implications of**

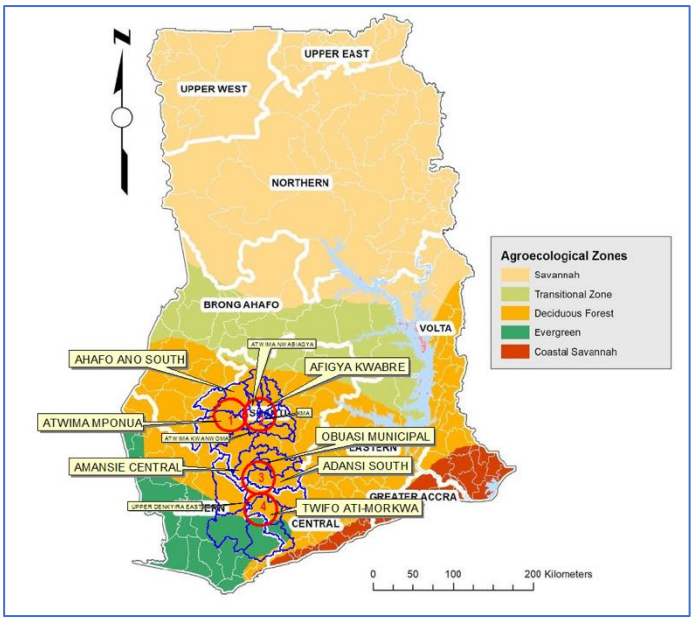


Figure 3: Map of Research Sites and Agroecological Zones. Prepared by Victor Addabor. Legend: districts are in blue, and study sites are circled in red

strategic partnerships for climate change, such as measuring the carbon stocks of different production systems and practices.

The southwestern part of Ghana represents today the center of cocoa production. Our study area will be located in the Ashanti and Central areas of the country, and located in forested areas of the country where more than 95% of cocoa producers are located (Figure 3) (Ghana Census of Agriculture). The districts covered by the study are all located in the deciduous forest agroecological zone to ensure a relative homogeneity in agroecological conditions (despite important forest cover variations). The northern area of study (1 and 2 on Figure 3), in the Ashanti region, comprises mainly the districts of Atwima Mponua, Ahafo Ano South, Afigya Kwabre, Kma, Atwima Nwabiagya, and small shares of other districts. Our southern area of study (3 and 4 on Figure 3) comprises the districts of Adansi South, Upper Denkyira East, Twifo Ati-Morkwa, Wassa East, and small shares of other districts. The rationale for the selection of the research sites is further developed in the Research Design section below.

Research design - District/forest comparative research design: We rely on a two-by-two dimensions research design that allows to maximize the comparison potential of our analysis along key dimensions: the level strategically-sourced cocoa and the influence of native forest/deforestation at the district-level for Q1 and Q2. **The goal of this research design will be to (1) control for important variations of ecosystem services at the district-level which eliminates the potential bias that companies have been strategically sourcing from areas of high native forest cover (2) control for past land-use history and how it may have affected ecological and institutional (e.g. trust among actors) variables in the district.**

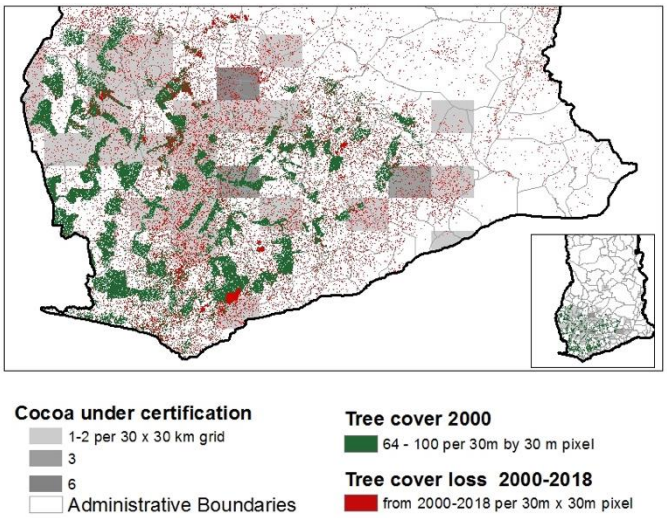


Figure 4: *Map of certified cocoa locations, forest cover, and deforestation.* ***Tree cover data****: Hansen et al. 2013*

We will characterize the level of strategically-sourced cocoa at the district-level using already available spatially-explicit data that shows the location of certified cocoa farms (each pixel is 20 x 20 km) (Tayleur et al. 2018) (Figure 4), and persistent tree cover and tree-cover loss (each pixel is 30 x 30 km) (Hansen et al. 2013a) (Version 1.6). These data will be summarized at the district level (Figure 3). We select two districts that present high levels of participation into strategic partnerships of strategically-sourced cocoa in the district) (>2 certifications per 20 by 20 km pixel), and two districts with low levels of participation into strategic partnerships (0% of strategically-sourced cocoa in the district). For each high and low strategic partnership site "pairs", one will be located in an area of poor tree cover (<20% of tree cover remaining in the district) and another will be located in an area of high tree cover (>65% of tree cover remaining in the district, based on Jenks natural breaks in the dataset). We treat the cocoa-district strategic partnership status (high or low fraction of strategic partnership labelled cocoa) as the treatment (or intervention) in a potential outcomes inference framework (Rubin 1974; 2005).

In each district ~30 farmers will be selected **randomly** based on a list of cocoa farmers provided by the local district authority (partnership secured by the Ghana Statistical Service). **We have confidence** that we can reach this number of surveyed farmers based on the previous experiences of our collaborator, Omar Seidu, at the Ghana Statistical Service, which manages the country's official agricultural census.

For Q2, we will use the same research design, but we will also make estimates at the national level using Landsat images across the country (4 scenes, with 2 essentially covering cocoa) and bar-coded data on the buyers and suppliers of cocoa at the district level. For Q3, we will be conducting financial models, and will use aggregated financial data for farmers, and corporates buying cocoa.

Methods and Analysis

Q1. Why do some farmers engage in strategic sourcing efforts, but not others? [Johnson, Van Holt, Hayek, Carter, Whelan]

Objective: To understand why not all farmers participate in strategic sourcing initiatives. Companies have invested in and rolled out strategic-sourcing initiatives across Ghana. Half of the cocoa is under some type of strategic sourcing. Here we disentangle the environmental, institutional, and network factors that may influence farmers' decisions to participate. We test our predictions on partner types:
Farmers who participate in strategic sourcing can be a logical partner, a peripheral partner, or a follower partner.

- **The logical partner:** H1: Farmers with strongly engaged strategic partners work on farms with a good environmental quality, have high trust scores in relation to others in their district, and have higher individual level social capital.
- **The peripheral partner:** H2: Farmers that have unique social network positions (structurally unique in terms of fewer network connections) may also have specialized knowledge and take more risks by entering in strongly engaged strategic partnerships regardless of the quality of their farm.
- **The follower partner:** H3: Farmers who have non-unique network positions (structurally embedded) also have their knowledge perceptions well-aligned with actors with which they are socially embedded. Their farms may be in a good or a poor quality area.

Data generation and synthesis overview: We will generate the following datasets: A ***strategic partnership engagement measure*** for each farmer to show their level of engagement with strategic sourcing. ***Ecosystem-profiles*** will be created for each farmer to signal how preserved or degraded the ecosystem is near their cocoa farms. Additional farmer-level surveys include a cultural consensus survey to generate information about institutional drivers (***legitimacy and trust***), and a social network analysis to understand additional socio-economic information (***position in social network***) that factors into farmer decisions of whether they engage in a strategic partnership.

[1.1.] Strategic partnership engagement [Carter,Whelan, and Van Holt]

Strategic-sourcing partnerships can take many forms ranging from only a price premium to full technical and financial assistance, and specific requirements to make farms more sustainable (Fenger et al. 2017, Gonockowski et al. 2013). Here we will survey the diversity of strategic partnership types as implemented on the ground by commodity firms and partnering farmers to develop a typology of partnerships (for H1). We will use a Guttman scale to operationalize the level of engagement (Guttman 1950). These data will be obtained at the district level, since all cocoa is brought in from various farms, and strategically sourced cocoa is separated and placed in bar-coded bags here. The bar-coded information contains the location, the buyer, and any special certifications that require traceability. This typology will recognize three dimensions to strategic partnerships: (1) <u>premium level,</u> farmers receive a premium-added price for their product that is higher than the market price; (2) sustainability <u>requirements,</u> farmers comply with sustainable agricultural, climate-friendly, practice requirements; (3) <u>assistance</u>, farmers may receive

technical assistance, financial assistance, or favorable contracting. First, we will code the presence or absence of such characteristics for each strategic partnership entered for each farmer of the study (especially since farmers may participate in several strategic partnerships). We will also cross-check this coding based on the official documents provided by the global commodity firms who established such partnerships. The final coding may be refined based on the evidence found in the field and may include also other favorable contracting conditions (e.g. long-term contracts, friendly terms of sale, supply chain finance, favorable terms of payment, and transparent pricing), as well as the request—e.g. organic farming conditions, reforestation stipulations, soil conditions, or labor stipulations. We will use strategic partnership engagement variables as a set of outcome variables stemming from our typology. In the pilot phase we will learn about the strategic partnership types, and then we will create a Guttman scale using those partnerships. We expect that low engagement would only include premiums, and high engagement would also include assistance, though this will be confirmed in the field.

[1.2.] Farm Ecosystem profiles [Hayek and Van Holt]
The objective of this analysis is to evaluate the ecosystem services existing on cocoa producer's farms, by synthesizing them in a farm ecosystem profile. The farm plot will be identified by relying on official land tenure data and confirmed in the field through land surveying with the farmers (collecting georeferenced point for the boundaries). The average cocoa farms in the study site being 2 to 5 hectares, we can reasonably assume that this can be collected for each farm of the sample during the field research phase. If a farmer has more than one plot, we will aggregate the plots together. We will rely on tree cover (including all trees, cocoa and shade trees) as a proxy for ecosystem services and draw on already existing forest cover data up to 2000 (Hansen et al. 2013b) or will conduct unsupervised classification of past land-uses relying on Landsat images for dates prior to 2000 (if needed).

[1.3] Farmer perceptions on cocoa production, the market, climate, and trust [Johnson, Carter, Van Holt]
The objective of this analysis is to isolate how farmer perceptions of the cocoa production, the market, climate and trust relate to an individual's social network. We will create two cultural consensus analysis surveys: one about farmer's environmental (including climate) and market knowledge and one about farmer's trust in their stakeholders (community, buyers, extension agents, etc.). To create these, there is a pilot phase where open-ended questions will be asked, a survey design phase where questions from the pilot survey will be transformed into a series of true/false statements about these domains. In phase 2, the cultural consensus will be administered in the field.

To construct the shared perceptions of cocoa production, the market, climate and trust, in-depth interviews will be used to develop a set of cultural statements reflecting culturally salient elements of these domains. Variation in shared perceptions and beliefs will be examined using the cultural consensus model (Romney, Weller, and Batchelder 1986). The cultural consensus model will allow for evaluation of the extent to which beliefs are shared across farmers. The patterns of responses the statements and will be put into an informant-by-proposition database that will allow for the statistical comparison across informants. Data in this form will allow for the application of both the formal and informal cultural consensus models (Romney, Weller, and Batchelder 1986). If there is a shared understanding of the beliefs, the ratio of the first to second eigenvalue in a minimum residual factor analysis will be greater than three and the scores on the first factor should all be positive with a mean score greater or equal to 0.5 (Romney, Weller, and Batchelder 1986). The consensus analysis, more importantly, allows for the means for determining how variation in beliefs is a function of actor characteristics (e.g., farming practices) and social network characteristics through a comparison of scores using multiple regression, ANOVA and MANOVA techniques (Boster and Johnson 1989; Johnson and Griffith 1996). Informant scores on the first factor of the minimum residual factor analysis are indicators of each informant's correlation to the aggregate understanding of cultural beliefs (e.g., challenges) to their cultural competency. The first factor score, the dependent variable, will be linearly

related to a series of quantitative and qualitative independent variables (e.g., network characteristics, education, etc.) in multivariate tests of hypotheses X-Y.

For the **cocoa production, market, and climate change knowledge**, we will aim for 20 environmental questions related to cocoa production, and 20 questions on the cocoa market, and 20 about climate change as it relates to cocoa. To elicit responses in the pilot phase, farmers will be asked grand tour, and mini tour questions (Johnson and Weller 20--). For the environmental questions, we will ask them to walk us through the process of planting, maintaining, and harvesting cocoa to understand how they know when to plant and thin trees, conduct pruning operations, spot and treat diseases and pests, whether the harvest season will be good or bad. For the market questions we will ask them questions such as when during the year do they make contracts, when is the high season, what factors affect price, what items do they negotiate for in the contracts, how do they coordinate the harvest, etc. For the climate change section, we will ask about changes in the growing season, timing of flowering, animal distribution, rainfall, or drought. For the **trust survey**, in phase 1, we will conduct open-ended interviews with farmers to ask them about cocoa production and marketing practices along with questions about the individuals and organizations they trust most for information related to these points. Then we will ask informants to think of something that they don't trust, and we will ask them to explain why is that so. Questions will also focus perceptions of legitimacy of different actors—NGOs, companies, the government, and other farmers. We then will develop a consensus analysis survey using the responses to the open-ended questions, and administer this in the field (phase 2). The consensus analysis for trust and legitimacy will be conducted as described above. This analysis will enable us to isolate if there is (or not) agreement on trust in terms of environmental issues or on supply chain issues, and examine how it affects levels of participation in strategic sourcing.

[1.4] Network position [Johnson, Van Holt, and Carter]
The objective of this analysis is to understand how a farmer's position in the network of actors involved in cocoa production and marketing may shape their cocoa production knowledge, and perceptions of trust. We will develop a whole-network study focused on cocoa farmer relationships for each of the study districts (Borgatti, Everett and Johnson 2017). We will survey the farmers in each of the four districts in the study. During phase 1, we will conduct interviews with key informants in a co-op community (prior to the surveying the farmers, the main participants of the study) to identify the names of the cocoa farmers in the co-op, as well as the name of government agents, leaders in the community, and cocoa supply chain actors. In phase 2, we will ask farmers to identify who they rely on for 1) environmental knowledge, and 2) market knowledge. While these are typically question for ego networks, given that each co-op contains about 30 farmers, we will use these data to generate ties. From this dataset, we will calculate network measures that identify unique structural positions including structural holes (which is a measure of social capital of ego), particularly betweenness centrality which will be used as an indicator of individual level social capital.

One of the important aspects of social capital is that it should get you something (R. S. Burt 2002). As Burt states, "The social capital metaphor is that people that do better are better connected" (202). At the group or societal level, social capital provides security, protection from out-group conflict, cooperation, and trust (Portes 1998; Coleman and Coleman 1994) whereas it provides individual actors with greater degrees of power, influence, brokerage, advantage, and monetary rewards. Social network characteristics at the group level (i.e., bonding social capital) involve higher levels of network density, redundancy, network closure, and greater network cohesion resulting in higher levels of trust and cooperation within a group at the expense of greater normative constraints and obligations. Social capital at the individual level (i.e., bridging social capital) involves networks of lower density, lower redundancy, with "ego" (around that individual) network structures that are extensive and complex. Individual actors connect to others who are not connected to one another providing access to novel information and other advantages as noted above. These networks provide fewer normative obligations and constraints, but they often imply greater social risks that can be diminished through an increase in social cohesion around ego (Tasselli et al. 2015). Barr, Ensminger, and Johnson (2009) found that those actors with higher levels of individual level social capital, as measured

using network betweenness centrality, were more trusting and trustworthy in trust experiments among the Orma of Kenya and workers in Accra, Ghana. The basic premise was that in order to build entrepreneurial social capital one has to risk trusting others in order to build a reputation for being trustworthy in repeated interactions (Burt 2002). The socio-political environment in our study area provides an interesting place to examine the role of social capital in the choice of farming practices.

Measuring Individual Level Social Capital: There are two primary ways to measure individual level social capital. As noted earlier,(Barr, Ensminger, and Johnson 2009) used betwenness centrality to measure social capital. Betweenness centrality is a path-based centrality measure that counts the number of times that a given node lies along the shortest path between two other nodes. Following (Freeman, Roeder, and Mulholland 1979) betweenness centrality will be determined as follows:

$$C_B(k) = 2\left(\sum_{i=1}^{n}\sum_{j=1}^{n}\left[\frac{g_{ij}(k)}{g_{ij}}\right]\right)$$

Here for all unordered triples *i, j, k* ($i<j$, and $i \neq j \neq k$), *n* is the number of nodes in the g_{ij} is the number of geodesics or shortest paths between node *i* and *j* and $g_{ik}(k)$ is the number of shortest paths from *i* to *j* that include *k*. We will also employ a number of other social capital measures discussed by (Portes 1998)in his theory of structural holes. These are ego based network measures and include constraint, effective size, and hierarchy. These measure are similar to betweenness centrality in that they measure the extent to which an ego (actor) is connected to a single set of actors (i.e., constrained) or to two or more sets of actors who are not connected to one another.

Structural Equivalence: In determining the structural uniqueness or embeddedness of a farmer in the social network, we will employ structural equivalence (Burt 1982). Two actors are structurally equivalent if they have the same ties to the same actors in the same way. Structurally unique actors are either highly central in the network or on the periphery of the network. The implications of this are that actors who have high degrees of overlapping ties will have higher degrees of shared beliefs, attitudes, and perceptions and will be more constrained by existing social norms (Follower Adopters). Conversely, structural unique actors, either very central or peripheral, are less constrained by social norms and may develop beliefs and perceptions that are more divergent (Logical Adopters and Peripheral Adopters). Johnson (1986) used structural equivalence to study the adoption of innovations among commercial fishermen. The study found that structurally unique actors with high centrality tended to adopt innovations earlier, while structurally embedded actors (highly overlapping ties) adopted innovations later and in quick succession as other structurally equivalent actors adopted the innovation. Thus, structurally unique central actors who have fewer social constraints were less subject to social norms that might impede early adoption. Structurally unique peripheral actors are also less socially constrained, but may adopt any time throughout the diffusion process. More structurally embedded actors face more normative constraints and therefor are likely to adopt the innovation quickly once other structurally equivalent actors adopt.

Analysis:

Once these different components gathered, we will test our hypotheses about farmer participation and producer profiles relying on **logistical regression analyses** where the dependent variable will be the participation of a producer in strategic sourcing. As stated above, the goal of this analysis will be to disentangle the respective influence of the environmental conditions (through the ecosystem-carbon profile of farm), institutional conditions. We will conduct preliminary data analyses by producing cross-tabulations of high vs low strategic partnership districts, and identifying the hypothesized underlying factors and for continuous variables simple Pearson correlations. Statistical tests for mean differences between the two groups with typical adjustments for multiple comparisons will provide a first indication of the associations.

A principal component analysis (PCA) will be done to verify that groupings of variables accurately describe the hypothesized farmer profiles. To determine the predictors (or causes) of cocoa producers engaged in high vs low levels of strategic partnerships (Q1), we will treat the partnership status as the dependent variable in a regression model with an appropriate link function (e.g. logit for binary outcomes). The relative importance of the different independent (explanatory) variables such as high legitimacy scores will be determined either by standardized regression coefficients or other appropriate methods (Grömping 2015).

Q2. Can strategic sourcing efforts increase climate resilience and farm productivity at scale? [Van Holt, Carter, Hayek, Johnson, Whelan]

Objective: To understand whether climate resilience and producer income (opportunity costs of carbon) can simultaneously increase as a result of strategic sourcing efforts, at scale. While we have some evidence of positive environmental outcomes from strategic sourcing in Ghana (Fenger et. al. 2017), we still have limited knowledge about whether the outcomes are a result of strategic sourcing efforts, or other factors. In drier parts of the cocoa belt of Ghana, for instance, farmers plant shade trees on their cocoa farms as a way to both adapt to climate change and provide an alternative source of income (Graefe et al. 2017). Ideally, strategic partnerships should provide more stable supply chains, and technical expertise that contributes to the adoption of sustainable agricultural practices among cocoa farmers. Such practices should lead to higher cocoa productivity and climate resilience. ***One of the debated challenges of sustainable intensification, however, is to what extent the adoption of climate-smart agricultural practices fosters or comes at the expense of productivity gains.*** Answering this question is crucial for understanding whether cocoa farms under strategic sourcing partnership represent a viable model for the sector that avoids the conversion of additional land to meet cocoa demand and satisfactorily address climate change risks.

Part 1: Carbon storage outcomes of strategic sourcing partnerships vary according to farmer type. First, we will draw from our logical, peripheral, and follower farmer types in Q1, and survey the same farms for **climate-smart farming techniques** and **productivity measures**. We will test which types of adopters can have both **climate resilience** and **higher productivity (income from farm)**.

- **Strategic partnerships and stored carbon:** H4: Producers engaged in strategic sourcing partnerships tend to have higher carbon stocks per unit of producer income than others.
- **The logical adopter:** H5: Likely have **higher** carbon stocks per unit of producer income because they have high trust scores in relation to others in their district, and have higher individual level social capital. Should have a good balance of high climate resilience and productivity, especially since they are located on farms with good environmental condition in the first place.
- **The peripheral adopter:** H6: Likely have **higher** carbon stocks per unit of producer income because they think and do things differently than others, and they seek out unique knowledge networks. The extent of climate resilience and productivity depends on how good the environmental condition was in the first place. They likely will adopt and keep low productivity—since climate resilience and productivity may be incompatible.
- **The follower adopter:** H7: Likely have **lower** carbon stocks per unit of producer income because they rest more on what others are doing than what the strategic partnership asks. Here there may be a problem with adoption of practices, and we may more of a focus on high productivity, rather than climate resilience, especially for those farms with low environmental quality in the first place.

Data generation and synthesis overview: We will generate the following datasets: An ***opportunity cost for carbon*** for each farmer. We will do this by surveying each farmer about their ***agricultural and forestry practices*** and modeling carbon stocks based on the diversity of practices, ***carbon estimates*** for each of those practices, and the ***farm productivity*** (income from all activities). We then calculate the carbon opportunity cost.

[2.1.] Productivity, income, and climate-smart practices [Hayek, Whelan, Van Holt, Johnson]
Here we will survey the 120 farms for general farm productivity and income information as well as evidence of climate friendly practices that may be mediating the stored carbon outcomes. This data-collection effort will go hand-in-hand since climate-smart practices may contribute to farm income. We will measure producer income based on **cocoa productivity** (tons of cocoa produced per hectare multiplied by the purchasing price) and other types of income received by the producer as a result of other agriculture-based activity linked to the farm. Climate-smart practices include (1) the presence of native varieties of non-cocoa shade trees on the farm; (2) the number of times farmer check for pests on cocoa trees; (3) the presence of non-cocoa trees producing income (diversification); (4) harvest regiments for secondary commercial non-cocoa trees (disturbance); (5) reforestation and soil conservation efforts; (6) watershed wide conversation measures-farmers engage in watershed or district-wide environmental improvements; (7) original land-clearing practices (corroborated through Hansen deforestation data and remotely-sensed land-use data as explained in Q1). We will evaluate the adoption of such practices during farm visits and validate the number of farm-level shade trees through our land-cover analysis (explained above). During the visit, we will also ask farmers under strategic sourcing partnerships to share with us written records of their practices.

[2.2.] Determining the opportunity costs of carbon for each farm [Hayek, Carter]
The objective of this analysis will be to model carbon stored (section 1.2) per unit of income generated at the farm-level (producer income). Our survey information of agricultural and forestry management practices and commercialization of section 2.1. will be matched to literature-reported values of below and above carbon (Gockowski and Sonwa 2011; Leuschner et al. 2013; Mohammed et al. 2016; Montagnini and Nair 2004; N'Gbala, Guéi, and Tondoh 2017). Literature values will provide estimates of carbon stocks, and a farmer surveys will inform estimates of the dynamics of these stocks over time. Carbon stock and dynamics information will be used to develop box models of cocoa production systems, used to produce estimates of time-averaged carbon stocks (TACS) on each farm. TACS is a reliable measure of the mean carbon storage over the lifetime of each farm through growth and timber harvesting dynamics. TACS will in turn inform our carbon opportunity cost analysis. This analysis will describe the benefits and tradeoffs of moving from one type of production system to another, or improving cocoa productivity within a given farm. Finally, we will compute carbon opportunity costs (COC) by calculating the total of producer income divided by TACS. This will help us identify the opportunity costs of carbon from one type of farm management to another, and, if strategic sourcing partnerships do influence sustainable management, therefore from one type of strategic sourcing partnership to another. We will also gather additional quantitative and qualitative measures of cocoa production (number of beans per pod, presence of fungus, harvest time, and profits) to test alternative outcome measures for our hypotheses.

Analysis: To test H4-H7, we will regress our dependent variable (carbon stored per unit of producer income) on the strategic sourcing variable, the different strategic partnership profiles that farmers are engaged in, and include contextual factors collected in the study (level of knowledge, trust, and network measures) that could act as mediators of the overall effect. The regression model analysis will consequently be expanded to investigate the putative mediating effects through interaction terms and a so-called mediation analysis.

Part 2. Scalability to support climate resilience and productivity across Ghana.
[Carter, Van Holt, Johnson, Hayek]

Then we will model whether these efforts can be made at scale. We will try to estimate whether multiple strategic partnerships can facilitate Ghana to reach their goal of 1 million cocoa plants and do so in a climate friendly way. We have an opportunity to test whether there is a combination of strategic partnerships at the district level that have an effect on the district-level carbon storage, and productivity potential. We will combine a behavioral network analysis of the ***Ghanaian cocoa supply network*** with ***landscape-level***

carbon, and ***productivity*** estimates. We will use farm-level data (on the 120 farms), to run a supervised classification across Ghana (Landsat 8 scenes) to create a carbon sequestration potential map centered on the cocoa sector (Hall et al. 2012). We will combine this with supply chain characteristics of the buyers to **test whether their supply chain network structures explain the potential to affect productivity and resilience at scale**.

- H8: Multiple strategic partnerships in a district provide synergistic effects (Akoyi and Maertens 2018), and this is associated with higher carbon potential

Data generation and synthesis overview: We will generate the following datasets: A model of the ***Ghanaian-wide cocoa supply network,*** and a ***landscape-level*** carbon analysis, and the s***ocio-economic demographics***. These data will be summarized at the district level. We will then estimate whether multiple strategic sourcing partnerships predict carbon storage potential, and identify whether different socio-economic benefits emerge.

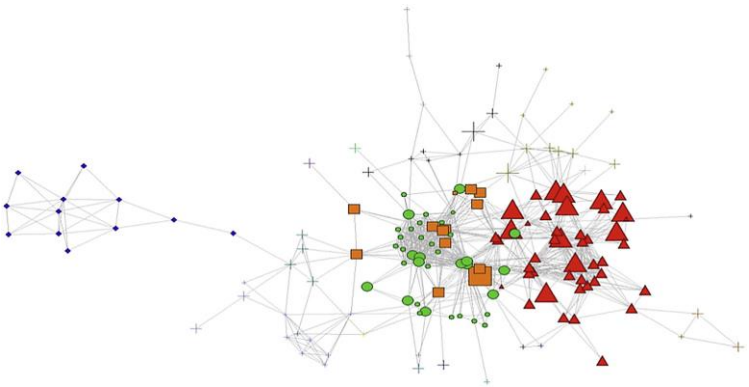


Figure 5 *Example of a hypothetical behavioral network where each node is a district, the size of the node may represent number strategic partnerships, and the colors represent districts that have similar strategic partnership strategies. We will discern if the pendant blue group for example, has lower carbon sequestration potential (few strategic partnerships) and the red group has many for example*

[2.3.] Ghanaian-wide cocoa supply network analysis

Behavioral supply networks will be created at each district level. To do this, we will use the Ghanaian bar-coded data on the entire cocoa supply chain. This classifies each bag of cocoa. This includes whether the cocoa comes from inside or outside of strategic partnerships, the source of the cocoa (at the district level), the amount of the cocoa, and the buyer. First, coco-sourcing communities were identified using a network. To do this, a district-by-buyer matrix, a district buying profile, was created where xij=1 if the ith district reported a specific buyer j and xij=0 otherwise. This will yield a two-mode matrix X. We can
look at the similarities in buyers across districts by producing a product matrix of X and the transpose XX′, where the ijth cell reflects the number of buyers that both districts i and j have in common.
This can be viewed as an indicator of similarities in sourcing behavior between each i and j. This yields an nxn one-mode district-by-district matrix (or network).
.

[2.4.] Landscape-level analysis [Van Holt]

We will calculate the cocoa farm types by creating a supervised classification of Landsat images. To do this we will first mosaic the most recently available, reliable, Landsat images for Ghana, and geographically and topographically correct the images (Landsat images 194/55 and 194/56). Prior to the field we will run an unsupervised classification, develop masks for specific land covers. For example, we will mask out the native forests isolate by the Hansen dataset, and also mask out clear bare soil/ crop agriculture, urban, and water classes. Our ground truthing will focus mainly on isolating the different types of cacao farms—especially isolating those with mixed forests from those with other crops interspersed. We will apply the standard number of ground truthing points for each land-cover class (minimum of 30 each and spatially distributed across the landscape). Of course, discerning between different types of cocoa farms is challenging, and we will collect many more ground truthing points for those land uses (Tracy Van Holt et

al. 2016). We will also collect another set of points to assess the accuracy of the image analysis. Carbon estimates will be related to land-use patterns and native forest classification according to Hall et al. (2012) to produce regional estimates of carbon changes over the entire southwestern Ghana domain. This will include using spatial statistics to model the carbon on the landscape. We will use the landscape-shape metrics in Hall et al. (2012) that influence native diversity at the landscape scale. This ranges from ''very high'' to ''negligible'' to indicate the potential of a landscape to support native biodiversity. Data on average carbon per hectare in each region were used to compute a spatially-weighted average in Mg C ha-1 for each sub-landscape. If it is too challenging to isolate different types of cocoa farms, we will rely on the 2018 agricultural census data in Ghana which identifies mono cropping and mixed cropping of the following: avocado, banana, cashew, cocoa, coconut, coffee, cola, citrus, mango, oil-palm, guava, papaw, shea nut and rubber.

[2.4.] Socio-economic data

We will rely on the 2018 agricultural census data in Ghana to identify the socio-economic variables of interest. We will also use this dataset to estimate farm productivity.

Analysis:

To find cohesive subsets of districts vis-a-vis their buyer, we will subject the behavioral network to the Girvan- Newman community detection algorithm (Girvan and Newman 2002). The algorithm finds cohesive sub communities using an edge betweenness removal approach. This will generate a district-buying profile for further analysis. We then will use a discriminate function analysis to identify whether the landscape level carbon estimates or socio-economic dataset explain the strategic partnership clusters. We can find out if there are synergies across strategic partnerships, and if they have the potential to address carbon storage.

Q3. What are the financial returns of increased climate resilience and productivity for the supply chain? (brands and farmers)? [Whelan and Van Holt]

Objective: To understand what business benefits, and the financial value of those benefits, emerge from investments to foster climate friendly, productive cocoa farms. Companies still struggle with the business case for sustainability (Reed 2001; Salzmann, Ionescu-Somers, and Steger 2005; Steger 2006). The concept of shared value (Porter and Kramer 2011) is prominent in the business literature but often plays a subordinate role. Moreover, the benefits of sustainability tend to enrich intangible assets, e.g. highly motivated employees, which corporate accounting is prone to miss (Lev 2000; Lev and Gu 2016). When stakeholders beyond shareholders are engaged, benefits can emerge that contribute to the financial success of a firm. Monetizing these benefits helps overcome one of the barriers of scaling up sustainability.

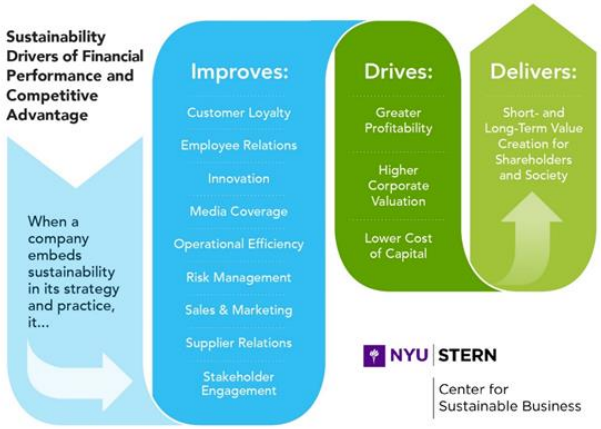


Figure 6 *The ROSI framework: Benefits emerge when companies embed sustainability. The drivers can be identified, quantified, and monetized to assess the financial value of sustainability actions.*

Data generation and synthesis overview: This analysis aims to quantify the financial benefits of sustainability investments, here related to the cocoa supply chain, for one or more supply chain companies. We propose to monetize the financial benefits of engaging in a strategic sourcing partnership in a case study

design. The NYU Stern Center for Sustainable (Stern CSB) Business Return on Sustainability Investment (ROSI) is a framework that monetizes the benefits of sustainability actions (Atz et al. 2019). We use a list of mediating factors that drive profits, avoid cost or reduce risk for a company (Figure 6). Non-financial accounting metrics attempt to summarize a company's position with regard to their earning potential, stakeholder engagement, environmental performance, and governance structure, but fall short in a variety of ways, e.g. measurement divergence (Berg et al. 2019), not focusing on material factors (Khan, Serafeim, and Yoon 2016) or simply a lack of detail. By getting access to internal metrics and estimates, we will be able to estimate the comprehensive financial value of a company's sustainability investments. Only if investors and other stakeholders are able to gain insights into how a company's investments in sustainability impact its financial performance, will there be a systemic change in how businesses improve supply chains in the long-term (Porter, Serafeim, and Kramer 2019).

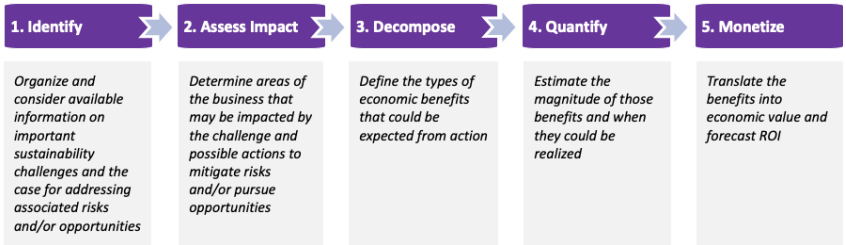


Figure 7 *The five steps in the Return of Sustainability Investment Methodology that are aimed at getting more precise estimates of the financial benefits of a sustainability initiative as compared to ESG metrics or traditional accounting.*

1.1. Monetizing the benefits of sustainability.

Analysis

The Stern CSB ROSI framework assesses the potential and realized benefits in terms of its short-term and long-term financial benefits. The short-term benefits often rely on accounting metrics such as earnings or cost savings that are netted against the capital and operating costs of the initiative. The latter, long-term view, typically takes into account the intangible benefits such as reduced risk of a supply chain disruption and are modelled with a net present value analysis.

In the first step of the ROSI methodology, we capture and elicit salient information regarding the benefits of strategic partnerships by organizing a series of semi-structured interviews with 6-10 participants from the case study company and 2-4 external stakeholders. These respondents are selected based on their familiarity with the project, seniority, and with at least one person representing the finance or accounting department.

We then, in step 2, determine a list that typically ranges from one to two dozen benefits pertaining to the sustainability investment or strategy. In this step, we also establish what historical data might be available to provide further basis for credible assumptions and estimates in the following steps. The process here is expected to be iterative as benefits from various sources may overlap and need to be refined into a concise and feasible set of impact areas.

Step 3 in turn defines the types of economic benefits as they are operationalized with concrete assumptions. A strategic partnership between a supplier and a corporation, for example, could involve a long-term contract, which employees may perceive as being more efficient. This efficiency gain then has to be broken down in manageable pieces such as 'saved days' and ideally corroborated by related initiatives where data already exists. Feedback is again requested from our project partners to validate modelling assumptions.

Step 4 gathers specific inputs, parameters, and data from the company and third-party sources. Inputs, for instance, relate to market estimates of the consumer demand for sustainably-sourced chocolate. Parameters are figures such as the discount rate, a value of time that a company may equate with their cost of capital. (Yet we found a practical range to be somewhere between 8-12%). The most critical piece are internal data that, if not centrally available, come from the procurement, marketing, and other departments within the company. We gather them in a second round of interviews and by providing a user-friendly spreadsheet that facilitates this process.

Finally, in step 5, the benefits and associated data are translated into monetary terms and estimated according to the most appropriate model. Appendix A in Atz et al. (2019) exhibits a detailed example and further complementary information is available as well. In our approach, the quantitative models are usually published and made available for free, which stands in stark contrast of the "black box algorithms" of financial consultancies and investment firms.

In a prior project (manuscript under review) we analyzed a specific sustainability program (step 1) for intermediaries that process coconuts and sit between farmers and global corporations. As described above, we applied the ROSI framework with the following results. In a series of interviews, stakeholders identified 22 business benefits (step 2) that we summarized into five benefit categories. To quantify and monetize the benefits we built an Excel model that operationalizes the assessment of the business benefits (step 3).[1] Data were supplied by the project partners and estimated from public sources (step 4). Lastly (step 5), we monetized the tangible and intangible (with a focus on the latter) benefits for one intermediary company. For example, the profit margin increase from a direct connection to farmers was estimated at $0.8-1.5 per metric ton of coconut.

Intellectual merit

Food companies around the world are exploring how best to create strategic partnerships that provide farmers with more equitable pricing and contracting standards that in turn help them implement practices reducing environmental impacts. The proposed research will fill an important gap by linking the content of such strategic partnerships to on-the-ground environmental outcomes, such as reforestation and sequestered carbon. The combination of different methods (network, cultural consensus, and land-use analyses) will help demonstrate the respective roles of local environmental conditions and community social networks in determining environmental outcomes and farmer participation. By modelling and monetizing the "carbon opportunity costs" and other sustainability benefits of different types of strategic partnerships and cocoa production systems for the first time, it will demonstrate the existence of win-win scenarios in which agricultural systems can sustain livelihoods and help mitigate climate change. The national-scale analysis linking supply chain characteristics to estimates of stored carbon will advance the understanding of climate change impacts from agroforestry systems. The benefits identified by the research project will inform the supply chain strategies of U.S.-based firms operating internationally and purchasing a large share of world commodities, as well as provide guidance to USAID policy. This research illustrates NSF's Growing Convergence Research idea, according to which the combination of disciplinary theories (forest transition, network, institutional theories) and methods can unlock key insights into the grand challenges of today.

Broader impact

This research project will inform policy and practice on how to structure sustainable supply chains and strategic sourcing partnerships for landscape- and farmer-level environmental and societal benefits. Driving this focus is the concern that if current problematic socio-economic, environmental and agronomic conditions are not addressed, companies will no longer be able to source cocoa from West Africa and farmers will no longer be able to make a living and support their families. Companies, despite hundreds of sustainability commitments, also still lack the necessary guidance to assess and scale up sustainability benefits. This project will offer critical insights to companies purchasing commodities and policy makers in charge of land-use policy, by identifying the key levers and impacts of strategic partnerships as well as monetary estimates of their benefits to help drive more investment.

This project will further interdisciplinary research by offering postdoctoral training based on merging disciplines (supply chain management, geography, ecology, and business) and research methods (network

[1] The model is available at: https://bit.ly/rosi-supplychains (Excel download)

analysis, remote sensing, and cultural consensus analysis). In addition, the project will engage a broad diversity of scholars engaged in CNH2 research by partnering with Ghana's Statistical Services and leveraging the NYU Accra satellite office. NYU Accra hosts NYU students, who come from a range of backgrounds. Educational opportunities will be provided through the development of a pedagogical case study that will be of interest to students of supply chains, business, and sustainability. NYU students in the U.S. will have the opportunity to participate in research on the project. We will also contribute to the infrastructure for further CHNS research though the activities of the NYU Center for Sustainable Business (CSB), which has a focus on applied research with companies. CSB aims to develop a science of strategic partnerships that helps define how sustainability provides a return on investment for all players in the value chain. This project builds on past research undertaken by CSB with large US retailers and food companies; CSB's director was the former president of the Rainforest Alliance (she will collaborate on the project).

Previous NSF support:
Johnson, JE.
NSF SPRF-Broadening Participation Award#:1450520 (2014-2019) $151,726. "Summer Course for Research Design in the Social, Behavioral and Economic Sciences (SCRD)". The course is co-directed by the PI Jeffrey Johnson (University of Florida) and Co-PI Christopher McCarty (University of Florida) and involves a number of other faculty including H. Russell Bernard (University of Florida), Kirk Johnson (University of Mississippi) and John Sonnett (University of Mississippi). The SCRD runs for one week each summer and was held on the campus of the University of Florida in Gainesville, Florida. The Summer Course for Research Design (SCRD) brings together social science area experts and post-doctoral participants from the social sciences for 1 week of intensive study of research design in the social, behavioral and economic sciences. The focus of the course was on the development of scientific research proposals. Criteria for selection of course participants included an applicant's interest in issues of research design and a commitment to advancing his or her proposal writing skills, as well as an interest in research relevant to underserved populations. Potential participants from under-represented groups were particularly encouraged to apply. The course promoted links to the SBE Postdoctoral Research Fellowship initiative at NSF (SPRF), which emphasizes broadening participation and interdisciplinary research in the behavioral and social sciences. Course content covered a broad spectrum of SBE research design issues, but with a heavy emphasis on the link between theory and methods and the importance of this link in writing well-conceived research proposals.

Collaborators and Advisors:
We will create an advisory board that will meet virtually biannually so the team can gain expert feedback and advice regarding the project. The board will include: A representative from industry who sources cocoa from Ghana, Jason Green, West Africa Director, Mars, Inc. A lead researcher who has expertise in research design, and who will be leading fieldwork in Ghana, Omar Seidu, Ghana Statistical Service. A full professor with extensive field-research focused on Ghanaian commodities, Yaw Nyarko, NYU. An NGO that works with cocoa farmers in Ghana and certifies cocoa, Han de Groot, CEO or an appropriate delegate, Rainforest Alliance. Additionally Omar Seidu will work with us to oversee data collection, and assist in providing other relevant datasets.

Mentoring of graduate and undergraduate students:
Graduate students and undergraduates will meet with the post-doctoral scholar weekly, and they will also present their work in the NYU-team's weekly meetings. They will be trained in how to conduct literature reviews, data management techniques, and be given specific tasks related to the project. They will be given a concise task for each semester, in addition to overall team support.

International research and collaborators

US relevance: US companies rely on cocoa from Ghana, who is among the top two producers. Our monetization work (see Q3) will identify the return on sustainability investments to companies, as well as for farmers. Our research shows that companies who do not invest in sustainability face reputational risks, and they also may face supply chain shortages. There are a number of other benefits, such as more resilient supply chains, and we will quantify how those efficiencies will benefit US companies. One of the US companies on our advisory board will be Mars, Inc., a major cocoa buyer and US company. Of course, consumers are demanding more ethically sourced products, as cocoa, and they too will benefit from research that can identify how to scale-up strategic sourcing so that it benefits farmers and mitigates climate change.

Local collaborators: We are including local scientists as collaborators, and they have helped in the design of this proposal. This includes Omar Seidu from the Ghana Statistical Service who works with Gabbey Kassim (cocoa specialist), and Victor Ohuruogu (GIS specialist who produced Figure 1). That team will work with us in the field. They will be essential in our pilot phase to make sure the questions are culturally appropriate, and reasonable to answer. Omar will provide a team of skilled interviewers who speak the local language (there are many languages in Ghana), and will help with the field logistics. In addition, Omar is on our advisory board, and will provide key insights throughout the project, interacting with the advisory board team. We plan to support a visiting scholar application for either Victor or Gabbey to visit the US and work on our data together. Our counterparts will be co-authors on manuscripts as appropriate—e.g. Victor would help in the GIS work, for example. We also will work with NYU Ghana and provide opportunities for students to intern on the project while they are in Ghana. Another advisory board member, Yaw Nyarko, is an NYU economics professor, and has extensive research in Ghana. We will also work with him in his advisory role capacity to make sure that we maximize our opportunities for collaboration.

REFERENCES

Akoyi, Kevin Teopista, and Miet Maertens. 2018. "Walk the Talk: Private Sustainability Standards in the Ugandan Coffee Sector." *The Journal of Development Studies* 54 (10): 1792–1818. https://doi.org/10.1080/00220388.2017.1327663.

Ashby, Alison. 2016. "From Global to Local: Reshoring for Sustainability." *Operations Management Research* 9 (3): 75–88. https://doi.org/10.1007/s12063-016-0117-9.

Atz, Ulrich, Tracy Van Holt, Elyse Douglas, and Tensie Whelan. 2019. "The Return on Sustainability Investment (ROSI): Monetizing Financial Benefits of Sustainability Actions in Companies." *Review of Business: Interdisciplinary Journal on Risk and Society* 39 (2): 1–31.

Baccini, A., W. Walker, L. Carvalho, M. Farina, D. Sulla-Menashe, and R. A. Houghton. 2017. "Tropical Forests Are a Net Carbon Source Based on Aboveground Measurements of Gain and Loss." *Science* 358 (6360): 230–34. https://doi.org/10.1126/science.aam5962.

Barr, Abigail, Jean Ensminger, and Jeffrey C Johnson. 2009. "Social Networks and Trust in Cross-Cultural Economic Experiments." *Whom Can We Trust*, 65–90.

Baumgart-Getz, Adam, Linda Prokopy, and Kristin Floress. 2012. "Why Farmers Adopt Best Management Practice in the United States: A Meta-Analysis of the Adoption Literature." *Journal of Environmental Management* 96: 17–25. http://dx.doi.org/10.1016/j.jenvman.2011.10.006.

Blackman, Allen, and Jorge Rivera. 2011. "Producer-Level Benefits of Sustainability Certification." *Conservation Biology* 25 (6): 1176–85. https://doi.org/10.1111/j.1523-1739.2011.01774.x.

Boster, James S, and Jeffrey C Johnson. 1989. "Form or Function: A Comparison of Expert and Novice Judgments of Similarity among Fish." *American Anthropologist* 91 (4): 866–889.

Bray, Joshua G., and Jeffrey Neilson. 2017. "Reviewing the Impacts of Coffee Certification Programmes on Smallholder Livelihoods." *International Journal of Biodiversity Science, Ecosystem Services & Management* 13 (1): 216–32. https://doi.org/10.1080/21513732.2017.1316520.

Burt, MG. 1992. "The Justification for Applying the Effective-Mass Approximation to Microstructures." *Journal of Physics: Condensed Matter* 4 (32): 6651.

Burt, Ronald S. 2002. "The Social Capital of Structural Holes." *The New Economic Sociology: Developments in an Emerging Field* 148: 90.

Burton, Rob J. F. 2004. "Reconceptualising the 'Behavioural Approach' in Agricultural Studies: A Socio-Psychological Perspective." *Journal of Rural Studies* 20 (3): 359–71. https://doi.org/10.1016/j.jrurstud.2003.12.001.

Carlisle, Liz. 2016. "Factors Influencing Farmer Adoption of Soil Health Practices in the United States: A Narrative Review." *Agroecology and Sustainable Food Systems* 40 (6): 583–613. https://doi.org/10.1080/21683565.2016.1156596.

Chiputwa, Brian, and Matin Qaim. 2016. "Sustainability Standards, Gender, and Nutrition among Smallholder Farmers in Uganda." *The Journal of Development Studies* 52 (9): 1241–57. https://doi.org/10.1080/00220388.2016.1156090.

Clough, Yann, Heiko Faust, and Teja Tscharntke. 2009. "Cacao Boom and Bust: Sustainability of Agroforests and Opportunities for Biodiversity Conservation." *CONSERVATION LETTERS* 2 (5): 197–205. https://doi.org/10.1111/j.1755-263X.2009.00072.x.

Coleman, James S., and James Samuel Coleman. 1994. *Foundations of Social Theory*. Harvard University Press.

Crona, Beatrice, Magnus Nyström, Carl Folke, and Narriman Jiddawi. 2010. "Middlemen, a Critical Social-Ecological Link in Coastal Communities of Kenya and Zanzibar." *Marine Policy*, Coping with global change in marine social-ecological systems, 34 (4): 761–71. https://doi.org/10.1016/j.marpol.2010.01.023.

DeFries, Ruth S., Jessica Fanzo, Pinki Mondal, Roseline Remans, and Stephen A. Wood. 2017. "Is Voluntary Certification of Tropical Agricultural Commodities Achieving Sustainability Goals for

Small-Scale Producers? A Review of the Evidence." *Environmental Research Letters* 12 (3): 033001. https://doi.org/10.1088/1748-9326/aa625e.

Fenger, Nina Astrid, Aske Skovmand Bosselmann, Richard Asare, and Andreas de Neergaard. 2017. "The Impact of Certification on the Natural and Financial Capitals of Ghanaian Cocoa Farmers." *Agroecology and Sustainable Food Systems* 41 (2): 143–66. https://doi.org/10.1080/21683565.2016.1258606.

Fountain, A. C., and F. Hütz-Adams. 2018. "Cocoa Barometer 2018."

Freeman, Linton C, Douglas Roeder, and Robert R Mulholland. 1979. "Centrality in Social Networks: II. Experimental Results." *Social Networks* 2 (2): 119–141.

Gardner, T. A., M. Benzie, J. Börner, E. Dawkins, S. Fick, R. Garrett, J. Godar, et al. 2019. "Transparency and Sustainability in Global Commodity Supply Chains." *World Development* 121 (September): 163–77. https://doi.org/10.1016/j.worlddev.2018.05.025.

Garrity, Dennis. 2012. "Agroforestry and the Future of Global Land Use." In *Agroforestry - The Future of Global Land Use*, edited by P.K. Ramachandran Nair and Dennis Garrity, 21–27. Advances in Agroforestry. Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-94-007-4676-3_6.

Girvan, Michelle, and Mark EJ Newman. 2002. "Community Structure in Social and Biological Networks." *Proceedings of the National Academy of Sciences* 99 (12): 7821–7826.

Glin, Laurent C., Peter Oosterveer, and Arthur P. J. Mol. 2015. "Governing the Organic Cocoa Network from Ghana: Towards Hybrid Governance Arrangements?" *Journal of Agrarian Change* 15 (1): 43–64. https://doi.org/10.1111/joac.12059.

Gockowski, James, Victor Afari-Sefa, Daniel Bruce Sarpong, Yaw B. Osei-Asare, and Nana Fredua Agyeman. 2013. "Improving the Productivity and Income of Ghanaian Cocoa Farmers While Maintaining Environmental Services: What Role for Certification?" *International Journal of Agricultural Sustainability* 11 (4): 331–46. https://doi.org/10.1080/14735903.2013.772714.

Gockowski, James, and Denis Sonwa. 2011. "Cocoa Intensification Scenarios and Their Predicted Impact on CO_2 Emissions, Biodiversity Conservation, and Rural Livelihoods in the Guinea Rain Forest of West Africa." *Environmental Management* 48 (2): 307–21. https://doi.org/10.1007/s00267-010-9602-3.

Graefe, Sophie, Lennart Flavio Meyer-Sand, Katja Chauvette, Issaka Abdulai, Laurence Jassogne, Philippe Vaast, and Richard Asare. 2017. "Evaluating Farmers' Knowledge of Shade Trees in Different Cocoa Agro-Ecological Zones in Ghana." *Human Ecology* 45 (3): 321–32. https://doi.org/10.1007/s10745-017-9899-0.

Grömping, Ulrike. 2015. "Variable Importance in Regression Models." *Wiley Interdisciplinary Reviews: Computational Statistics* 7 (2): 137–152.

Guttman, L.A. 1950. *The Basis for Scalogram Analysis. In Measurement and Prediction. Volume 4 of Studies in Social Psychology*. Princeton: Princeton University Press.

Hall, J.M., T. van Holt, A.E. Daniels, V. Balthazar, and E.F. Lambin. 2012. "Trade-Offs between Tree Cover, Carbon Storage and Floristic Biodiversity in Reforesting Landscapes." *Landscape Ecology* 27 (8). https://doi.org/10.1007/s10980-012-9755-y.

Hansen, M. C., P. V. Potapov, R. Moore, M. Hancher, S. A. Turubanova, A. Tyukavina, D. Thau, et al. 2013a. "High-Resolution Global Maps of 21st-Century Forest Cover Change." *Science*. https://doi.org/10.1126/science.1244693.

Hansen, M. C., P. V. Potapov, R. Moore, M. Hancher, S. A. Turubanova, A. Tyukavina, D. Thau, et al. 2013b. "High-Resolution Global Maps of 21st-Century Forest Cover Change." *Science*. https://doi.org/10.1126/science.1244693.

Johnson, Jeffrey C, and David C Griffith. 1996. "Pollution, Food Safety, and the Distribution of Knowledge." *Human Ecology* 24 (1): 87–108.

Khan, Mozaffar, George Serafeim, and Aaron Yoon. 2016. "Corporate Sustainability: First Evidence on Materiality." *The Accounting Review* 91 (6): 1697–1724. https://doi.org/10.2308/accr-51383.

Ki, Fiona Cheung Yan, and Steve Rowlinson. 2011. "Supply Chain Sustainability: A Relationship Management Approach." *International Journal of Managing Projects in Business* 4 (3): 480–97. https://doi.org/10.1108/17538371111144184.

Kuhfuss, Laure, Raphaële Préget, Sophie Thoyer, and Nick Hanley. 2016. "Nudging Farmers to Enrol Land into Agri-Environmental Schemes: The Role of a Collective Bonus." *European Review of Agricultural Economics* 43 (4): 609–36. https://doi.org/10.1093/erae/jbv031.

Kuyah, Shem, Cory W. Whitney, Mattias Jonsson, Gudeta W. Sileshi, Ingrid Öborn, Catherine W. Muthuri, and Eike Luedeling. 2019. "Agroforestry Delivers a Win-Win Solution for Ecosystem Services in Sub-Saharan Africa. A Meta-Analysis." *Agronomy for Sustainable Development* 39 (5): 47. https://doi.org/10.1007/s13593-019-0589-8.

Läderach, P., A. Martinez-Valle, G. Schroth, and N. Castro. 2013. "Predicting the Future Climatic Suitability for Cocoa Farming of the World's Leading Producer Countries, Ghana and Côte d'Ivoire." *Climatic Change*. https://doi.org/10.1007/s10584-013-0774-8.

Lambin, Eric F., Holly K. Gibbs, Robert Heilmayr, Kimberly M. Carlson, Leonardo C. Fleck, Rachael D. Garrett, Yann le Polain de Waroux, et al. 2018. "The Role of Supply-Chain Initiatives in Reducing Deforestation." *Nature Climate Change* 8 (2): 109–16. https://doi.org/10.1038/s41558-017-0061-1.

Larsen, Rasmus Kløcker, Maria Osbeck, Elena Dawkins, Heidi Tuhkanen, Ha Nguyen, Agus Nugroho, Toby Alan Gardner, Zulfahm, and Paul Wolvekamp. 2018. "Hybrid Governance in Agricultural Commodity Chains: Insights from Implementation of 'No Deforestation, No Peat, No Exploitation'' (NDPE) Policies in the Oil Palm Industry.'" *Journal of Cleaner Production*. https://doi.org/10.1016/j.jclepro.2018.02.125.

Leuschner, Christoph, Gerald Moser, Dietrich Hertel, Stefan Erasmi, Daniela Leitner, Heike Culmsee, Bernhard Schuldt, and Luitgard Schwendenmann. 2013. "Conversion of Tropical Moist Forest into Cacao Agroforest: Consequences for Carbon Pools and Annual C Sequestration." *Agroforestry Systems* 87 (5): 1173–87. https://doi.org/10.1007/s10457-013-9628-7.

Lev, Baruch. 2000. *Intangibles: Management, Measurement, and Reporting*. Brookings Institution Press.

Lev, Baruch, and Feng Gu. 2016. *The End of Accounting and the Path Forward for Investors and Managers*. John Wiley & Sons.

Meyfroidt, Patrick, Thomas K. Rudel, and Eric F. Lambin. 2010. "Forest Transitions, Trade, and the Global Displacement of Land Use." *Proceedings of the National Academy of Sciences of the United States of America*. https://doi.org/10.1073/pnas.1014773107.

Mills, Jane, Peter Gaskell, Julie Ingram, Janet Dwyer, Matt Reed, and Christopher Short. 2017. "Engaging Farmers in Environmental Management through a Better Understanding of Behaviour." *Agriculture and Human Values* 34 (2): 283–99. https://doi.org/10.1007/s10460-016-9705-4.

Mohammed, Askia M., James S. Robinson, David Midmore, and Anne Verhoef. 2016. "Carbon Storage in Ghanaian Cocoa Ecosystems." *Carbon Balance and Management* 11 (1): 6. https://doi.org/10.1186/s13021-016-0045-x.

Montagnini, F., and P. K. R. Nair. 2004. "Carbon Sequestration: An Underexploited Environmental Benefit of Agroforestry Systems." *New Vistas in Agroforestry*, 281–95. https://doi.org/10.1007/978-94-017-2424-1_20.

Munasib, Abdul B.A., and Jeffrey L. Jordan. 2011. "The Effect of Social Capital on the Choice to Use Sustainable Agricultural Practices." *Journal of Agricultural and Applied Economics* 43 (2): 213–27. https://doi.org/10.22004/ag.econ.104619.

N'Gbala, Fulbert N'Guessa, Arnauth Martinez Guéi, and Jérôme Ebagnerin Tondoh. 2017. "Carbon Stocks in Selected Tree Plantations, as Compared with Semi-Deciduous Forests in Centre-West Côte d'Ivoire." *Agriculture, Ecosystems & Environment* 239 (February): 30–37. https://doi.org/10.1016/j.agee.2017.01.015.

Ordonez, Jenny C, Eike Luedeling, Roeland Kindt, Hesti Lestari Tata, Degi Harja, Ramni Jamnadass, and Meine van Noordwijk. 2014. "Constraints and Opportunities for Tree Diversity Management

along the Forest Transition Curve to Achieve Multifunctional Agriculture." *Current Opinion in Environmental Sustainability*, Sustainability challenges, 6 (February): 54–60. https://doi.org/10.1016/j.cosust.2013.10.009.

Porter, Michael E., and Mark R. Kramer. 2011. "Creating Shared Value." *Harvard Business Review* 89 (Janury): 1–13.

Porter, Michael E, George Serafeim, and Mark Kramer. 2019. *Where ESG Fails*.

Portes, Alejandro. 1998. "Social Capital: Its Origins and Applications in Modern Sociology." *Annual Review of Sociology* 24 (1): 1–24.

Prokopy, L., K. Floress, D. Klotthor-Weinkauf, and A. Baumgart-Getz. 2008. "Determinants of Agricultural Best Management Practice Adoption: Evidence from the Literature." *Journal of Soil and Water Conservation* 63 (September): 300–311. https://doi.org/10.2489/jswc.63.5.300.

Putz, Francis E., and Kent H. Redford. 2010. "The Importance of Defining 'Forest': Tropical Forest Degradation, Deforestation, Long-Term Phase Shifts, and Further Transitions." *Biotropica*. https://doi.org/10.1111/j.1744-7429.2009.00567.x.

Reed, Donald. 2001. *Stalking the Elusive Business Case for Corporate Sustainability*. Washinghton: World resources Institute. https://www.worldcat.org/title/stalking-the-elusive-business-case-for-corporate-sustainability/oclc/875646874&referer=brief_results.

Ribaudo, Marc. 2015. "The Limits of Voluntary Conservation Programs." *Choices: The Magazine of Food, Farm, and Resource Issues* 30 (2): 1–5.

Romney, A Kimball, Susan C Weller, and William H Batchelder. 1986. "Culture as Consensus: A Theory of Culture and Informant Accuracy." *American Anthropologist* 88 (2): 313–338.

Rubin, Donald B. 1974. "Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies." *Journal of Educational Psychology* 66 (5): 688.

———. 2005. "Causal Inference Using Potential Outcomes: Design, Modeling, Decisions." *Journal of the American Statistical Association* 100 (469): 322–331.

Rudel, Thomas K., Oliver T. Coomes, Emilio Moran, Frederic Achard, Arild Angelsen, Jianchu Xu, and Eric Lambin. 2005. "Forest Transitions: Towards a Global Understanding of Land Use Change." *Global Environmental Change*. https://doi.org/10.1016/j.gloenvcha.2004.11.001.

Rudel, Thomas K., Patrick Meyfroidt, Robin Chazdon, Frans Bongers, Sean Sloan, H. Ricardo Grau, Tracy Van Holt, and Laura Schneider. 2019. "Whither the Forest Transition? Climate Change, Policy Responses, and Redistributed Forests in the Twenty-First Century." *Ambio*, January. https://doi.org/10.1007/s13280-018-01143-0.

Ruf, François, Götz Schroth, and Kone Doffangui. 2015. "Climate Change, Cocoa Migrations and Deforestation in West Africa: What Does the Past Tell Us about the Future?" *Sustainability Science* 10 (1): 101–11. https://doi.org/10.1007/s11625-014-0282-4.

Salzmann, Oliver, Aileen Ionescu-Somers, and Ulrich Steger. 2005. "The Business Case for Corporate Sustainability: Literature Review and Research Options." *European Management Journal* 23 (1): 27–36. https://doi.org/10.1016/j.emj.2004.12.007.

Searchinger, Timothy D., Stefan Wirsenius, Tim Beringer, and Patrice Dumas. 2018. "Assessing the Efficiency of Changes in Land Use for Mitigating Climate Change." *Nature* 564 (7735): 249–53. https://doi.org/10.1038/s41586-018-0757-z.

Sloan, Sean, Patrick Meyfroidt, Thomas K. Rudel, Frans Bongers, and Robin Chazdon. 2019. "The Forest Transformation: Planted Tree Cover and Regional Dynamics of Tree Gains and Losses." *Global Environmental Change* 59 (November): 101988. https://doi.org/10.1016/J.GLOENVCHA.2019.101988.

Smith, B. Gail. 2008. "Developing Sustainable Food Supply Chains." *Philosophical Transactions of the Royal Society B: Biological Sciences* 363 (1492): 849–61. https://doi.org/10.1098/rstb.2007.2187.

Spash, Clive L., Kevin Urama, Rob Burton, Wendy Kenyon, Peter Shannon, and Gary Hill. 2009. "Motives behind Willingness to Pay for Improving Biodiversity in a Water Ecosystem: Economics, Ethics and Social Psychology." *Ecological Economics*, Participation and Evaluation

for Sustainable River Basin Governance, 68 (4): 955–64. https://doi.org/10.1016/j.ecolecon.2006.09.013.

Steger, Ulrich. 2006. "Building a Business Case for Corporate Sustainability." *Managing the Business Case for Sustainability: The Integration of Social, Environmental and Economic Performance*, 412–443.

Streck, Charlotte, Alan Kroeger, Simon Koenig, and Ashley Thomson. 2017. "Forest, and Climate-Smart Cocoa in Côte D'Ivoire and Ghana : Aligning Stakeholders to Support Smallholders in Deforestation Free-Cocoa." 122086. The World Bank. http://documents.worldbank.org/curated/en/317701513577699790/Forest-and-climate-smart-cocoa-in-C%C3%B4te-D-Ivoire-and-Ghana-aligning-stakeholders-to-support-smallholders-in-deforestation-free-cocoa.

Tasselli, Stefano, Ronald S. Burt, Martin J. Kilduff, Adam M. Kleinbaum, Alessandro Lomi, Ajay Mehra, Marco Tortoriello, and Daniel Feiler. 2015. "Exploring the Micro-Foundations of Organizational Social Networks." *Academy of Management Proceedings* 2015 (1): 12264. https://doi.org/10.5465/ambpp.2015.12264symposium.

Tayleur, Catherine, Andrew Balmford, Graeme M. Buchanan, Stuart H.M. Butchart, Christine Corlet Walker, Heather Ducharme, Rhys E. Green, et al. 2018. "Where Are Commodity Crops Certified, and What Does It Mean for Conservation and Poverty Alleviation?" *Biological Conservation* 217 (January): 36–46. https://doi.org/10.1016/J.BIOCON.2017.09.024.

Tscharntke, Teja, Jeffrey C. Milder, Gotz Schroth, Yann Clough, Fabrice DeClerck, Anthony Waldron, Robert Rice, and Jaboury Ghazoul. 2015. "Conserving Biodiversity Through Certification of Tropical Agroforestry Crops at Local and Landscape Scales." *Conservation Letters*, no. 1: 14. https://doi.org/10.1111/conl.12110.

Van Holt, Tracy, Wendy R. Townsend, and Peter Cronkleton. 2010. "Assessing Local Knowledge of Game Abundance and Persistence of Hunting Livelihoods in the Bolivian Amazon Using Consensus Analysis." *Human Ecology* 38 (6): 791–801. https://doi.org/10.1007/s10745-010-9354-y.

Van Holt, T. 2012. "Landscape Influences on Fisher Success: Adaptation Strategies in Closed and Open Access Fisheries in Southern Chile." *Ecology and Society* 17 (1). https://doi.org/10.5751/ES-04608-170128.

Van Holt, Tracy, Harvey Bernard, Susan Weller, Wendy Townsend, and Peter Cronkleton. 2016. "Influence of the Expert Effect on Cultural Models." *Human Dimensions of Wildlife* 21 (2): 169–79. https://doi.org/10.1080/10871209.2015.1110736.

Van Holt, Tracy, Michael W. Binford, Kenneth M. Portier, and Rodrigo Vergara. 2016. "A Stand of Trees Does Not a Forest Make: Tree Plantations and Forest Transitions." *Land Use Policy*. https://doi.org/10.1016/j.landusepol.2016.04.015.

Van Holt, Tracy, and Wendy Weisman. 2016. "Global Production Network Mapping for Transforming Socio-Ecological Systems." *Current Opinion in Environmental Sustainability*, Sustainability challenges, 20 (June): 61–66. https://doi.org/10.1016/j.cosust.2016.06.003.

Van Holt, Tracy, B. Crona, J.C. Johnson, and S. Gelcich. 2017. "The Consequences of Landscape Change on Fishing Strategies." *Science of the Total Environment* 579. https://doi.org/10.1016/j.scitotenv.2016.10.052.

Van Holt, Tracy, and F.E. Putz. 2017. "Perpetuating the Myth of the Return of Native Forests." *Science Advances* 3 (5). https://doi.org/10.1126/sciadv.1601768.

Weisse, Mikaela J., and Elizabeth Dow Goldman. 2019. "The World Lost a Belgium-Sized Area of Primary Rainforests Last Year." *World Resources Institute* (blog). April 25, 2019. https://www.wri.org/blog/2019/04/world-lost-belgium-sized-area-primary-rainforests-last-year.

West, Paul C., Holly K. Gibbs, Chad Monfreda, John Wagner, Carol C. Barford, Stephen R. Carpenter, and Jonathan A. Foley. 2010. "Trading Carbon for Food: Global Comparison of Carbon Stocks vs. Crop Yields on Agricultural Land." *Proceedings of the National Academy of Sciences* 107 (46): 19645–48. https://doi.org/10.1073/pnas.1011078107.

Williams, David R., Ben Phalan, Claire Feniuk, Rhys E. Green, and Andrew Balmford. 2018. "Carbon Storage and Land-Use Strategies in Agricultural Landscapes across Three Continents." *Current Biology* 28 (15): 2500-2505.e4. https://doi.org/10.1016/j.cub.2018.05.087.

Winters, Paul, Hsuan-Wen Kuo, Chanisa Niljinda, Ben Chen, Helena Nery Alves-Pinto, Melisa Ongun, Stefani Daryanto, and Peter Newton. 2015. "Voluntary Certification Design Choices Influence Producer Participation, Stakeholder Acceptance, and Environmental Sustainability in Commodity Agriculture Sectors in Tropical Forest Landscapes." *Journal of Sustainable Forestry* 34 (6–7): 581–604. https://doi.org/10.1080/10549811.2015.1017884.

Wossen, Tesfamicheal, Thomas Berger, Teferi Mequaninte, and Bamlaku Alamirew. 2013. "Social Network Effects on the Adoption of Sustainable Natural Resource Management Practices in Ethiopia." *International Journal of Sustainable Development & World Ecology* 20 (6): 477–83. https://doi.org/10.1080/13504509.2013.856048.

Yoder, Landon, Adam S. Ward, Kajsa Dalrymple, Scott Spak, and Rebecca Lave. 2019. "An Analysis of Conservation Practice Adoption Studies in Agricultural Human-Natural Systems." *Journal of Environmental Management* 236 (April): 490–98. https://doi.org/10.1016/j.jenvman.2019.02.009.

Youn, Sunhee, Ma Ga (Mark) Yang, Paul Hong, and Kihyun Park. 2013. "Strategic Supply Chain Partnership, Environmental Supply Chain Management Practices, and Performance Outcomes: An Empirical Study of Korean Firms." *Journal of Cleaner Production*, Sustainability management beyond corporate boundaries, 56 (October): 121–30. https://doi.org/10.1016/j.jclepro.2011.09.026.

Yu, Yang, Kuishuang Feng, and Klaus Hubacek. 2013. "Tele-Connecting Local Consumption to Global Land Use." *Global Environmental Change* 23 (5): 1178–86. https://doi.org/10.1016/j.gloenvcha.2013.04.006.

Zimmerer, Karl, Eric Lambin, and Steven Vanek. 2018. "Smallholder Telecoupling and Potential Sustainability." *Ecology and Society* 23 (1). https://doi.org/10.5751/ES-09935-230130.

Tracy Van Holt
Director of Academic Research
NYU Stern Center for Sustainable Business
40 West 4th Street
New York, NY 10012
tvanholt@stern.nyu.edu
www.tracyvanholt.com

(a) Professional Preparation

SUNY at Stony Brook	Stony Brook, NY	Biochemistry	B.S.	1994
University of Florida	Gainesville, FL	Interdisciplinary Ecology	M.S.	2001
University of Florida	Gainesville, FL	Interdisciplinary Ecology	Ph.D.	2009
East Carolina University	Greenville, NC	Data mining / Network analysis	Post Doc	2009

(b) Appointments

- Director of Academic Research, NYU Stern Center for Sustainable Business. 2017-present.
- Senior Researcher at the Royal Swedish Academy of Sciences in the Global Economic. Dynamics and the Biosphere Programme, Stockholm, Sweden, 2014-2017.
- Marine Theme Co-Leader, Stockholm Resilience Center, Stockholm, Sweden. 2014-2017.
- Assistant Professor in Geography and the Institute for Coastal Science and Policy, East Carolina University, Greenville, NC. 2010-2014.

(c) Publications/products

(i) Most closely related to the proposed project (5)

1. Atz, U., **Van Holt, T**., Douglas, E., and T. Whelan. 2019. Monetizing sustainability actions in global companies to understand their potential financial benefits. Review of Business. Interdisciplinary Journal on Risk and Society. 39(2):1-31.
2. **Van Holt, T.**, Crona, B., Johnson, J.C., and S. Gelcich. 2017. The consequences of landscape change on fishing strategies. Science of the Total Environment. 59:930-939. http://dx.doi.org/10.1016/j.scitotenv.2016.10.052
3. **Van Holt, T.**, Binford, M.W., Portier, K.M., and R. Vergara. 2016. A stand of trees does not a forest make: Tree plantations and forest transitions. Land Use Policy. 56:147-157. 10.1016/j.landusepol.2016.04.015
4. **Van Holt, T.**, Bernard, H.R., Weller, S. Townsend, W., and P. Cronkleton. 2016. The expert effect in cultural models. Human Dimensions of Wildlife Management. 21(2):169-179. 10.1080/10871209.2015.1110736
5. **Van Holt, T.** 2012. Landscape influences on fisher success: adaptation strategies in closed and open access fisheries in southern Chile. Ecology and Society 17(1):28. http://dx.doi.org/10.5751/ES-04608-170128

(ii) Other significant products (5)

1. Crona, B., Käll, S., and T. Van Holt. 2019. Fishery Improvement Projects as a governance tool for fisheries sustainability: A global comparative analysis. PLoS ONE 14(10): e0223054. https://doi.org/10.1371/journal.pone.0223054
2. Rudel, T.K., Meyfroidt, P., Chazdon, R., Bongers, F., Sloan, S., Grau, R., Van Holt, T., and L. Schneider. 2019. Whither the forest transition? Climate change, policy responses, and redistributed forests in the twenty-first century. Ambio (2019). https://doi.org/10.1007/s13280-018-01143-0

3. Gordon, L., Bignet, V., Crona, B., Henriksson, P.J.G., Van Holt, T., et al. 2017. Rewiring food systems to enhance human health and biosphere stewardship. Environmental Research Letters 12(10). DOI10.1088/1748-9326/aa81dc
4. Crona, B., Van Holt, T., Petersson, M., Daw, T.M., and E. Buchary. 2015. Using social-ecological syndromes to understand impacts of international seafood trade on small-scale fisheries. Global Environmental Change. 25:162-175. doi:10.1016/j.gloenvcha.2015.07.006
5. Hall, J.M., Van Holt, T., Daniels, A.E., Balthazar, V., and E.F. Lambin. 2012. Trade-offs between carbon storage and floristic biodiversity in reforesting landscape. Landscape Ecology 27(8):1135-1147. http://dx.doi.org/10.1007/s10980-012-9755-y

d) Synergistic Activities (5)

1. Co-led workshops with industry: Building the business case for sustainability, H&M and Accenture Global Change Winners, May 2018.
2. Co-led workshop with Transformations in Global Fisheries. January 28-29, 2015, Royal Swedish Academy of Sciences, Stockholm, Sweden.
3. Co-led and developed and taught distance learning course on: Geospatial Analysis for Cultural University of Florida and Arizona State University 2012-2020
4. Co-developed an analysis to quantify agreement in participatory mapping.
5. Co-developing methods to quantify the return on sustainability investments.

Craig R. Carter
John G. and Barbara A. Bebbling Professor of Supply Chain Management
300 E Lemon St, Tempe, AZ 85281
craig.carter@asu.edu

(a) Professional Preparation

University of Maryland	College Park, MD	Logistics	B.S. 1990
Arizona State University	Tempe, AZ	Supply Chain Mgt.	Ph.D. 1996

(b) Appointments

Professor of Supply Chain Management, W.P. Carey School of Business, Supply Chain Management Department, Arizona State University, 2013 to Present

Associate Professor of Supply Chain Management, W.P. Carey School of Business, Supply Chain Management Department, Arizona State University, 2011 to 2013.

Associate Professor of Supply Chain Management and International Business, Managerial Sciences Department, College of Business Administration, University of Nevada, 2006 to 2011.

Assistant Professor of Supply Chain Management and International Business, Managerial Sciences Department, College of Business Administration, University of Nevada, 2004 to 2005.

Assistant Professor of International Supply Chain Management, Department of Logistics, Business & Public Policy, The Smith School of Business, University of Maryland, 1997 to 2003.

(c) Publications/products

(i) Most closely related to the proposed project (5)

1. Rogers, Zachary S., Craig R. Carter, and Virginia Kwan (2019), "Making Tough Choices: A Policy Capturing Approach to Evaluating the Tradeoffs in Sustainable Supplier development Initiatives," *Journal of Purchasing and Supply Management*, 25(5), forthcoming.
2. Wichmann, Barbara K., Craig R. Carter, and Lutz Kaufmann (2016), "Making Environmental SCM Initiatives Work – Moving Beyond the Dyad to Gain Affective Commitment," *Journal of Supply Chain Management*, 52(1), 21-40.
3. Ehrgott, Matthias, Felix Reimann, Lutz Kaufmann, and Craig R. Carter (2011), "Social Sustainability in Selecting Emerging Economy Suppliers," *Journal of Business Ethics*, 98(1), pp. 99-119.
4. Carter, Craig R., and Dale S. Rogers (2008), "A Framework of Sustainable Supply Chain Management: Moving toward New Theory," *International Journal of Physical Distribution and Logistics Management*, 38(5), pp. 360-387.
5. Carter, Craig R. (2005), "Purchasing Social Responsibility and Firm Performance: The Key Mediating Roles of Organizational Learning and Supplier Performance," *International Journal of Physical Distribution and Logistics Management*, 35(3), pp. 177-194.

(ii) Other significant publications (5)

1. Kaufmann, Lutz, Joerg Rottenburger, Craig R. Carter, and Christian Schlereth (2018), "Bluffs, Lies, and Consequences: A Reconceptualization of Bluffing in Buyer-Supplier Negotiations," *Journal of Supply Chain Management*, 54(2), 49-70.

2. Wichmann, Barbara K., Craig R. Carter, and Lutz Kaufmann (2015), "How to Become Central in an Informal Social Network: An Investigation of the Antecedents to Network Centrality in an Environmental SCM Initiative," *Journal of Business Logistics*, 36(1), 1-18.
3. Gattiker, Thomas F., Craig R. Carter, Wendy L. Tate, and Xiaowen Huang (2014), "Managerial Commitment to Sustainable Supply Chain Management Projects," *Journal of Business Logistics*, 35(4).
4. Gattiker, Thomas F. and Craig R. Carter (2010), "Understanding Project Champions' Ability to Gain Intra-Organizational Commitment for Environmental Management Projects," *Journal of Operations Management*, 28(1), pp. 72-85.
5. Pullman, Mellie, Michael J. Maloni, and Craig R. Carter, (2009) "Food for Thought: Motivations to Adopt Sustainability Practices and Perceived Outcomes," *Journal of Supply Chain Management*, 45(4), pp. 38-54.

d) Synergistic Activities (5)

1. Served as the Co-Editor-in-Chief of the *Journal of Supply Chain Management* (2007-2016)
2. Serve as an Associate Editor for *Journal of Operations Management* (2012-Present)
3. Serve as an Editorial Reviewer for *Sustainability* (2018-Present)
4. Serve as an Associate Editor for Journal of Business Logistics (2010-Present)
5. Serve as an Associate Editor for *Decision Sciences* (2015-Present)

Matthew N. Hayek
New York University

Professional Preparation

Wesleyan University, Middletown, CT	Physics	BA, 2007
Harvard University, Cambridge, MA	Environmental Engineering	PhD, 2017
Harvard University, Cambridge, MA	Law and Policy	Post-doc, 2018

Appointments

Assistant Professor	New York University Dept. of Environmental Studies	2019-Present
Postdoctoral Researcher	Harvard Animal Law and Policy Program	2017-2019
Postdoctoral Research	Harvard John A. Paulson SEAS	2017
Graduate RA	Harvard John A. Paulson SEAS	2009-2017

Publications/Products

Related Publications (5)

1. **Hayek, M. N.** & Garrett, R. D. Nationwide shift to grass-fed beef requires larger cattle population. *Environ. Res. Lett.* **13**, 084005 (2018).
2. **Hayek, M. N.** *et al.* Carbon exchange in an Amazon forest: from hours to years. *Biogeosciences* **15**, 4833–4848 (2018).
3. **Hayek, M. N.** *et al.* A novel correction for biases in forest eddy covariance carbon balance. *Agric. For. Meteorol.* **250–251**, 90–101 (2018).
4. Wu, J., Guan, K., **Hayek M. N.** *et al.* Partitioning controls on Amazon forest photosynthesis between environmental and biotic factors at hourly to interannual timescales. *Glob. Chang. Biol.* **23**, 1240–1257 (2017).
5. Miller, S. M., **Hayek, M. N.**, Andrews, A. E., Fung, I. & Liu, J. Biases in atmospheric CO2 estimates from correlated meteorology modeling errors. *Atmos. Chem. Phys.* **15**, 2903–2914 (2015).

Additional publications (3)

1. Wu, J., Albert, L., Aline, P., Restrepo, N., **Hayek, M.**, *et al.* Leaf development and demography explain photosynthetic seasonality in Amazon evergreen forests. *Science.* **351**, 972–977 (2016).
2. **Hayek, M. N.** *Drivers of Long-Term Variability in Amazon Forest Carbon Fluxes*. Harvard University doctoral dissertation (2017).
3. Longo M., Knox R. G., Levine N. M., Alves L. F., Bonal D., Camargo P. B., Fitzjarrald D. R., **Hayek M. N.**, *et al.* (2018) Ecosystem heterogeneity and diversity mitigate Amazon forest resilience to frequent extreme droughts. *New Phytologist*, 219(3):914–931. https://doi.org/10.1111/nph.15185.

Synergistic Activities (4)

1. **Agriculture, forests, and land use.** Ongoing research related to national and global agricultural land and resource demands. This research engages a multi-disciplinary team in food systems, land use, and ecology in scenario modeling exercises of land demand in the present and future. Modeling of IPCC 21st century emissions pathways to constrain upper and lower bounds of ecosystem carbon sequestration contributions towards emissions mitigation targets.
2. **National and Global management gaps in pulse crop yields.** Active research program involving quantifying geospatial distribution of resource management and yield attainment in pulse crop production. Past and current work involves using a statistical crop-climate modeling approach to quantify shortfalls between present and attainable yields conditional upon climate and resource

inputs. Current research is being performed in collaboration with researchers at land grant universities. This research also brings together interest in the nonprofit and market research sectors to provide input and economic analysis regarding the the provision, processing and manufacture of greater quantities of pulse ingredients. Research outputs will also inform agricultural development research collaborations in adaptation and mitigation option, such as those being explored by the agricultural model inter-comparison and improvement project (AgMIP). Improved pulse protein yields will inform projects on adapting food systems to climate change for enhanced food security, augmenting ongoing research regarding development of Climate Smart Agriculture (CSA) strategies.

3. **Understanding determinants and impacts of crop substitutions to realize social and environmental benefits.** Conducting data synthesis, statistical analysis, and scenario modeling to assess the drivers and impacts of crop substitutions nationwide at the US county level. The aim of this research is to understand how farmers in the United States (US) adapt to changing physical, economic, and social environments, such as climate change, market shocks, and dietary shifts, and to understand the ecological and social outcomes of these responses. In particular we aim to assess: (1) past and present environmental and socioeconomic impacts of cropping decisions; (2) economic, social, and environmental drivers that prompt farmers to substitute crops over time; (3) consequences of changing crop practices under scenarios of climate change, growing global demand for meat, and increasing trade barriers. Results of this project will be used to engage agricultural professionals and inter-institutional food systems experts in evaluating possible policy levers to encourage more socially equitable and environmentally sustainable crop practices and food systems.
4. **Mentorship at NYU.** I am responsible for the advising of undergraduate students majoring in Environmental Studies at NYU. I will be responsible for the advising of 1-3 undergraduate for their senior thesis projects, many of which relate to food systems and environmental sustainability. Additionally, I will be overseeing masters' students' capstone projects in the areas of agricultural sustainability and agroforestry management.

Jeffrey C Johnson
johnsonje@ufl.edu

Professional Preparation

1975 B.A. Anthropology, University of California, Irvine

1981 Ph.D. Social Science, University of California, Irvine

Appointments

2016-present-University Term Professor
2014-present-Preeminent Professor of Anthropology, University of Florida
2011-present-Adjunct Professor, Institute for Software Research, Carnegie Mellon University
2010-2013-Social Science Program Manager, Army Research Office (IPA)
2010-2014-Thomas Harriot College of Arts and Sciences Distinguished Professor (ECU)
2007-2014-University Distinguished Research Professor (ECU)
2007-2014-Senior Scientist, Institute for Coastal Science and Policy (ECU)
1997-2014-Professor, Department of Sociology (ECU)
1996-2014-Senior Scientist, Institute for Coastal and Marine Resources (ECU)
2002-2014-Adjunct Professor, Department of Biology (ECU)
1996-2014 -Adjunct Professor, Department of Biostatistics and Anthropology (ECU)
1995-1997-Associate Professor, Department of Sociology (ECU)
1986-1995-Associate Scientist, Institute for Coastal and Marine Resources (ECU)
1993-1994-Associate Professor, Department of Biostatistics (ECU)
1989-1992 Associate Professor, Department of Sociology and Anthropology (ECU)
1986-1989 Assistant Professor, Department of Sociology and Anthropology (ECU)
1981-1986 Assistant Scientist, Institute for Coastal and Marine Resources (East Carolina University)

Publications/products

(i) Most closely related to the proposed project (5)

1. Van Holt, T., B. Crona, J. C. Johnson, S. Gelcich. The consequences of landscape change on fisher-harvesting behavior. Science of the Total Environment. Volume 579, 1 February 2017, Pages 930-939.
2. S. Borgatti, M. Everett, and J. C. Johnson. Analyzing Social Networks. Second Edition. Sage Publications, London, (2017).
3. W. G. Ambrose Jr., L. M. Clough, J. C. Johnson, M. Greenacre, D. Griffith, M. L. Carroll and Alex Whiting. Interpreting Environmental Change in Coastal Alaska Using Traditional and Scientific Ecological Knowledge. Frontiers in Marine Science **1**:40. doi: 10.3389/fmars.2014.00040.
4. J.C. Johnson, J. Luczkovich, S.P. Borgatti and T. Snijders. Using Social Network Analysis Tools in Ecology: Markov Process Transition Models Applied to the Seasonal Trophic Network Dynamics of the Chesapeake Bay. Ecological Modeling, 220 (2009) 3133–3140.
5. J.C. Johnson and D.C. Griffith. "Finding Common Ground in the Commons: Intra-Cultural Variation in Users' Conceptions of Coastal Fisheries Issues". Society and Natural Resources, Vol.23:1-19, 2010.

(ii) Other significant publications (5)

1. Dell'Apa, A, J.C. Johnson, D. G. Kimmel, R.A. Rulifson. The international trade and fishery management of spiny dogfish: A social network approach. Ocean and Coastal Management 80 (2013): 65-72.
2. J. C. Johnson, R. R. Christian, J. W. Brunt, C. R. Hickman, and R. B. Waide. Evolution of Collaboration within the US Long-term Ecological Research Network, BioScience, Dec 2010 : Volume 60 Issue 1: 931-940.
3. J.C. Johnson and M.K. Orbach. "Perceiving the Political Landscape: Ego Biases in Cognitive Political Networks". Social Networks 24 (2002) 291-310.
4. J.C. Johnson, J.S. Boster, and D. Holbert. "Estimating Relational Attributes From Snowball Samples Through Simulation," Social Networks 11:135-158, 1989.
5. J.C. Johnson. "Social Networks and Innovation Adoption: A Look at Burt's Use of Structural Equivalence," Social Networks 8:343-364, 1986.

Synergistic Activities (5)

1. Co-Chair and Chair of the 2014, 2015 and 2016 International Social Computing, Behavioral Modeling and Prediction Conference (SBP), Washington DC.
2. Member, National Academies of Science Committee on the Decadal Survey of Social and Behavioral Sciences for Applications to National Security. (2017-present)
3. Member, National Academies of Science Committee on "Organizational Modeling from Individuals to Societies" (2004-2008)
4. Member, National Academies of Science Committee on "Marine Recreational Fishing Information Program Review" (2016-2018)
5. Faculty, LINKS Network Conference 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018 Gatton College of Business and Economics, University of Kentucky.

New York University Stern School of Business: Facilities, Equipment and Other Resources

New York University (NYU) is one of the nation's largest private institutions of higher education, with an enrollment of approximately 48,000 students. The University has 12 schools, colleges and divisions. It is one of only 27 private institutions affiliated with the Association of American Universities. Students attending the University come from all 50 states and 110 foreign countries to study at five centers in Manhattan. Over 2,500 courses are offered, leading to over 235 different degrees.

New York University maintains seven libraries containing over three million volumes. The main library, the Elmer Holmes Bobst Library and Study Center, is one of the largest open stack research libraries in the country. The library has 2.5 million books and journals, plus microforms, video and audio-tapes, other materials located in open stack. A computerized catalog provides access to most of the libraries holdings. The University also has six specialized libraries that contain more than 168,000 volumes, 2,000 periodicals, and computer software and audiovisuals.

The NYU Television Center, which also acts as a resource, manages satellite downlinks for instructional and educational programming, and provides on-campus teleconferencing.

High Performance Computing at NYU and NYU Stern School of Business

Stern supports a small computing cluster for researchers. Sun Grid Engine is used to schedule jobs on the cluster. The cluster has approximately 65 cores spread across 10 heterogeneous machines. More than 30 terabytes of data storage is available in the Stern cluster.

In addition to the Stern cluster, NYU Information Technology Services (ITS) supports high performance computing and networking for researchers and scholars, and is home to several high performance clusters and high-speed networks equipped with a wide variety of research software packages. Use of ITS high performance computing resources is open to all faculty and to sponsored graduate students.

The NYU General Cluster is a general purpose high performance computing resource for the NYU research community that offers 1.28 terabytes of memory, large-scale storage options, and a theoretical peak performance of 10 Teraflops (a trillion calculations per second). The Rocks operating system is used to manage the cluster, which consists of 1140 cores. Home directories are mounted from the control node of the cluster, and there is also a scratch filesystem (11 terabytes) accessible from all nodes.

NYU's IBM eServer BladeCenter system (Max) is capable of a peak performance of 5 teraflops (one trillion floating point operations per second). The acquisition of this supercomputer, which supports NYU research with heavy computational requirements, was made possible by a gift from IBM and federal funding. Storage on Max is 8.7 terabytes.

A new cluster, Cardiac, has recently been implemented. Funded principally from foundation funding to the School of Medicine, an investment by ITS allowed for the acquisition of additional cores in order to serve the NYU community. The cluster has 320 cores available for the NYU community and 960 cores for the researchers who received the funding.

New York University Accra (Ghana)

Each semester NYU Accra welcomes 20-40 students from one of NYU's degree-granting campuses (New York, Abu Dhabi and Shanghai). The program also offers opportunities for deep immersion in the society through its community service and academic internship programs both of which allow students to engage with a range of agencies, institutions and organizations.

The NYU Academic Center is equipped with a small library and a computer lab. Outside of the Center, researchers have access to the University of Ghana-Legon Balme Library with its rare African studies book collection, the W. E. B. Du Bois Memorial Centre for Pan-African Culture, and the National Museum, to name a few.

Unpaid Collaborators

Tensie Whelan will serve as an advisor to corporate liaisons, and will be in charge of supporting research activities regarding the cocoa supply chain network mapping as well as developing outreach

activities at the Stern Center for Sustainable Business. Her tasks will also include Broader Impacts activities of the NSF project. This includes furthering the work of NYU's Stern Center for Sustainable Business on corporate strategic partnerships, already well-established the Center and Whelan's past work with large US retailers and food companies. She will work to disseminate the findings of the project to global business conferences, and in management publications such as the Harvard Business Review weighing in on debates about corporate sustainability.

Data Management Plan

Roles and responsibilities for the project team:

- "The *Principal Investigator* is the custodian for the University of Research Data, including the scientific data and other records of research conducted at or under the auspices of the University" in accordance with the Policy on Retention of and Access to Research Data. Should the PI leave the project, we will follow NYU Guidelines. If a Principal Investigator leaves the University, and a project is to be moved to another institution, original Research Data may be transferred with the approval of the Vice Provost for Research (or a designee), and with written agreement from the PI's new institution that guarantees: 1) its acceptance of custodial responsibilities for the data, and 2) the University's access to the data, should that become necessary)."[1]
- All *professional researchers* on the team will have access to Data specified below. Only team members that completed the Collaborative Institutional Training Initiative (CITI) training for human subject research will have access to human subject data.
- *Postdoctoral Researchers and Students* will have access to the Data specified below with individual allowances made for Data that is under NDA or pertains to human subjects. For human subjects data, the respondent's name and other personally identifiable information will be stored in separate locations from the de-identified data.
- *Industry Project Partners*' role is primarily to support the project by providing data sets and guidance.

Data summary (further details are in the proposal):

Spatial data

- Spatially-explicit data from public sources, as well as the Ghanaian statistics authority
- GPS points and locations of farms
- Open data (e.g. satellite images)

Environmental data

- Soil analysis results
- Biodiversity transect survey results.

Field data on individual farmers (human subjects data)

- Notes, recording, and videos from the field (these will be stored with respondents information de-identified and located in a separate folder).
- Preliminary survey information--transcripts, and coding of data from interviews.
- Survey instruments for estimating traditional knowledge, trust and legitimacy.
- Social network survey instrument with farmers.
- Field experiment survey responses (vignettes)

Industry data

- Transcripts from calls with industry experts and coding of these data
- Any proprietary financial company data or otherwise will have an NDA agreement

Metadata

- Each data set will have respective metadata associated with it such as date of collection. This is especially useful for GIS data because it facilitates data sharing.

Data Management, Storage, and Retention

The PI holds original Research Data in accord with the NYU policy and the following specifications. Sufficient records will be maintained such that:

[1] https://www.nyu.edu/about/policies-guidelines-compliance/policies-and-guidelines/retention-of-and-access-to-research-data.html

- Field work data will be collected and stored on electronic devices that are password protected and connected the internet (if available) for cloud storage.
- CSB aims to maintain three copies of Research Data: (1) stored locally on the PI's machine, (2) stored on a secondary machine maintained by a CSB Researchers in a different physical location, and (3) stored on a cloud service provider such as NYU Drive[2] or NYU Box.[3] Where this is not feasible for size issues, either two copies will be stored or the cloud facility provides a dedicated backup system (see High Performance Computing)
- Notes will be stored for all team members on NYU Drive.
- A system of data organization will set clear conventions among team members on file naming and dates following the NYU Data Services guidelines.[4] Open source software such as R[5] and non-proprietary file types such as Comma-Separated Values (CSV) will be used except in circumstances where this is not possible.
- We will communicate this policy and the chosen system of data organization to all members of the project team.
- Original data will be maintained beyond the end of the project including, but not limited to, mandated guidelines such as the retention for six years following final project close-out.

Sharing and Dissemination

- Commercial and noncommercial data sharing agreements will be obtained before the launch of the project with the industry project partners. These include provisions for publications.
- Data sharing with the public and as part of publications follows standard practice in academia. Data sets will be made publicly available, as long as they are sufficiently anonymized and adhere to NDA agreements. The preferred repository is Harvard's Dataverse.[6]
- Computational models and code for non-proprietary analyses will be stored in a private GitHub repository and made available to the public within one year after the completion of the project.
- In general, we comply with the National Science Foundation (NSF) Grant Policy Manual (NSF 05-131) for dissemination and sharing of research results.

High Performance Computing

For complex operations such as potentially arising from the analyses of imaging data, CSB collaborates with the NYU Stern Center for Research Computing (SCRC) which provides world class computational facilities and services to researchers at Stern. The SCRC supports a moderate size grid of computers, which allow researchers to attack problems that are computationally challenging or that require large amounts of disk storage. The architecture of the Stern GRID is such that it can be dynamically extended without researchers having to modify their jobs. The SCRC has its own dedicated servers and backup facilities. An overview of the hardware, software, and database services can be found on their webpage.[7]

[2] https://www.nyu.edu/life/information-technology/websites-storage-and-sharing/nyu-drive.html
[3] https://www.nyu.edu/life/information-technology/websites-storage-and-sharing/nyu-box.html
[4] https://guides.nyu.edu/data_management/data-creation
[5] https://www.r-project.org/
[6] https://www.re3data.org/repository/r3d100010051
[7] http://scrc.scrc.nyu.edu/scrc/

Postdoctoral Researcher and Student Mentoring Plan

Scope: This proposal includes collaboration and mentoring for project involvement of a (1) Postdoctoral Researcher for the duration of the project and (2) NYU Stern School of Business Graduate and Undergraduate Students for the duration of two or more semesters. CSB has hosted, supported, and worked in close collaboration with five postdoctoral researchers / visiting scholars and 15 students in a research capacity since its launch in 2016. In addition, CSB is part of the Stern Program for Undergraduate Research (SPUR)[1] which connects NYU Stern students with research opportunities outside of the classroom. Finally, NYU Ghana mentors around 20 undergraduate students in study abroad programs, some of them may participate in an internship related with this project in Ghana.

Orientation: Before and during the project, orientation will include in-depth conversations between the PI, CSB researchers, and the Postdoctoral Researcher. Mutual expectations will be discussed and agreed upon. Orientation topics will include (a) the amount of independence the Postdoctoral Researcher requires, (b) interaction with coworkers, (c) productivity including the importance of scientific publications, (d) work habits and safety for field work, and (e) documentation of research methodologies and field details so that the work can be continued by other researchers in the future.

Career Development: Counseling will provide the Postdoctoral Researcher and Students with the skills, knowledge, and experience needed to excel in their chosen career path. The Postdoctoral Researcher will be trained on how to manage an interdisciplinary team of scientists. They gain skills in collecting and analyzing environmental, social, and spatial data. They will be exposed to NYU's class on non-traditional academics in business. Finally, CSB over 250 Stern Alumni working in sustainability related fields.

Research Management: The Postdoctoral Researcher will have an opportunity to learn best practices in proposal preparation including identification of key research questions, definition of objectives, description of approach and rationale, and construction of a work plan, timeline, and budget. They will also learn about interdisciplinary, and international, cross-cultural management.

Publications and Presentations: These will be prepared in collaboration with PI and team members as appropriate. The Postdoctoral Researcher will receive guidance and training in the preparation of manuscripts and conference presentations for multiple disciplines—business, sociology, environmental studies, and geography. They will be taught the nuances of interdisciplinary publications.

Teaching and Mentoring Skills: The Postdoctoral Researcher will have an opportunity to guest lecture in NYU classes, and will mentor MBA and SPUR students. The project will have regular meetings where the Postdoctoral Researchers will describe their work to student team members. We will regularly reflect on how to create an environment to teach and mentor so as to provide solutions to research challenges.

Instruction in Professional Practices: The Postdoctoral Researcher will receive foundational training in human subjects research and the NYU Institutional Review Board (IRB).

Technology Transfer: The Postdoctoral Researcher will be given an opportunity to become familiar with the university-industry relationships. This includes applicable confidentiality requirements and disclosure applications. Moreover, the Postdoctoral Researcher will gain insights into the operations of one of the largest global food supply networks from an academic and industry perspective.

[1] https://www.stern.nyu.edu/portal-partners/current-students/undergraduate/academics/research-practitioner-learning/spur

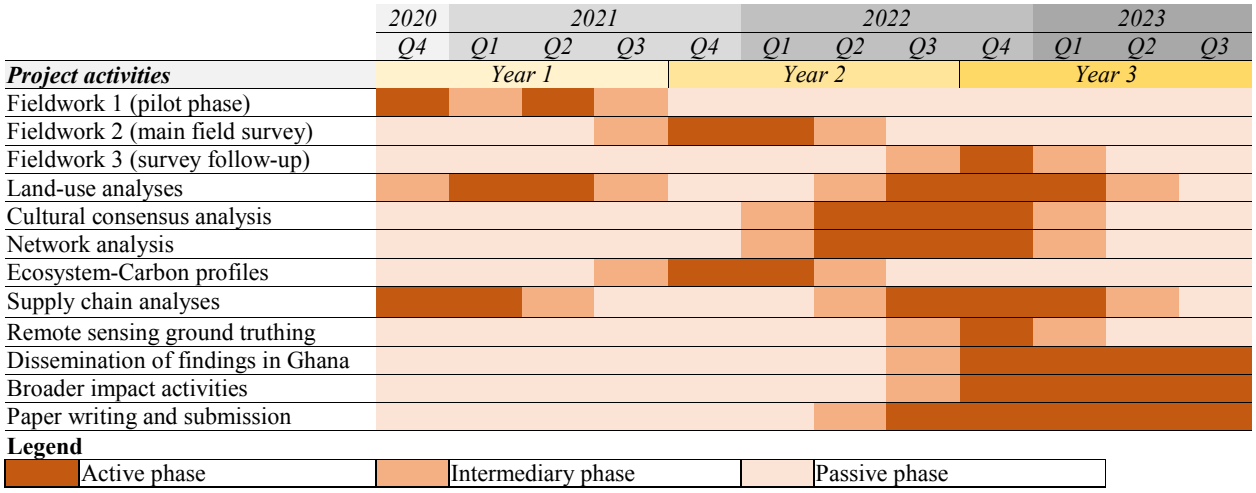
Project Management and Timeline

The projected activities will start in September 2020 and extend throughout August 2023. We will be structuring the key activities of the project in three phases corresponding to the project years. For the first two years of the project, our project timeline is considerably influenced by the fact that we need to align our fieldwork and interview with the season of cocoa in Ghana which extends from September to December.

Project team management

The overall project will be overseen by PI Tracy Van Holt. The PI, co-PIs, postdoctoral researcher, and research assistants will participate in bi-weekly meetings to manage the different tasks of the project and planned the following activities. The postdoctoral researcher will be in charge of facilitating the logistics of such meetings as well as planning the three yearly project meetings to be scheduled at the end of each project year (corresponding to 1 yearly meeting before fieldwork, 1 yearly meeting after fieldwork and the midst of data analysis phases, and finally 1 meeting during the broader impacts activities and paper submission phases). Land use and cultural consensus analyses will be overseen by Van Holt, network analysis by Johnson, ecosystem carbon profiles by Hayek, supply chain analyses by Carter, and broader impact activities will be overseen by Whelan. Team members associated with each variable are identified within each section. There will be a champion for each analysis, but overall we will discuss the results and analyses in our team meetings.

Detailed project schedule



Year 1 (September 2020 to August 2021)

During year 1 we will conduct two fieldwork trips: a first trip during Q4 (2020) for the pilot phase of the project where we will develop the survey instrument and a second trip Q2 (2021) to test the survey instrument in the field and collect ground truthing points for the remote sensing analysis (participants: PI Van Holt, co-PI Johnson, postdoctoral researcher). During that year we will also conduct the land-use analysis necessary to build out the landscape variables important for site selection, ecosystem profiles, and carbon stocks analysis. Finally, we will gather remotely and locally data about cocoa supply chain structures by making data requests to public agencies (such as the Cocoa Board of Ghana), licensed COCOBOD purchasing companies, and cooperatives in order to map out the supply chain. Finally, we will start fieldwork activities after preparation during Q3 of 20212.

Year 2 (September 2021 to August 2022)

During the first half of year 2 we will carry out field survey in Ghana over 4 months (from September 2021 to January 2022), aligning with the cocoa growing season when farmers can be interviewed (participants: postdoctoral researcher for the whole duration, co-PIs for selected periods). During fieldwork, our local research team will interview farmers and create ecosystem-carbon profiles of farms, under the supervision of the postdoctoral researcher and short visits of co-PIs involved in different steps of the data collection. During fieldwork we will complete data-entry tasks into the project database daily. The second half of year 2 will be spent on the project analyses, namely the cultural consensus survey, network, and supply chain analyses. We will also begin the country-wide carbon storage analysis for the cocoa supply chain.

Year 3 (September 2022 to August 2023)

Over the first half of year 3, the project team will finish the three main analyses of the project and continue the country-wide carbon storage analysis for the cocoa supply chain. We will undertake our third and last fieldwork trip to Ghana for two weeks in order to collect ground truthing points for the land-use analysis, visit the four different sites of the study to share the results of the field survey with local participants, and collect any missing data as needed as part of a field-survey follow-up. During this year we will also carry out other activities **as detailed in our Broader Impacts section, including conferences, outreach, and teaching activities**. The PI, Co-PIs, and postdoctoral researcher will also write up the different papers associated with the project and submit them to the top journals in their respective field.

MARS WRIGLEY

To: Whom it May Concern

National Science Foundation
Dynamics of Integrated Socio-Environmental Systems Program
2415 Eisenhower Avenue, Alexandria, VA 2231

Mars, Incorporated
Floor 7
No 4 Kingdom Street
Paddington
W2 6BD

November 15, 2019

Dear Sirs,

If the proposal submitted by Dr. Tracy Van Holt entitled "Transforming tropical commodity supply chains to mitigate climate change and foster sustainable livelihoods" is selected for funding by NSF, it is my intent to collaborate and/or commit resources as detailed in the Project Description or the Facilities, Equipment and Other Resources section of the proposal.

Yours sincerely,

Jason Green
Regional Director Cocoa, West Africa.

+225 05958604
jason.green@effem.com

mars.com

better moments make the world smile

Mars, Incorporated
Paddington, London
United Kingdom
Mars.com

© 2019 Mars, Incorporated

Tel: 0302682683
Cables: **GHANASTATS**

In case of reply the number and date of this letter should be quoted.

Our Ref. No. GSS/SDGs/14/11/19/07
Your Ref. No……..

REPUBLIC OF GHANA
STATISTICAL SERVICE
P. O. BOX GP 1098
ACCRA

14th November, 2019

COUPLED HUMAN NATURAL SYSTEMS
NATIONAL SCIENCE FOUNDATION
ALEXANDRIA
VIRGINIA
USA

Dear Sir/Madam,

LETTER OF SUPPORT FOR THE PROPOSAL ON TRANSFORMING TROPICAL COMMODITY CHAINS FOR MITIGATING CLIMATE CHANGE AND SUSTAINABLE LIVELIHOODS

I write to support the proposal submitted by Dr Van Holt entitled "Transforming tropical commodity supply chains for mitigating climate change and sustainable livelihoods". If selected for funding by National Science Foundation's Coupled Human Natural Systems, it is my intent to collaborate and/or commit resources as detailed in the Project Description section of the proposal.

Thank you,

Omar Seidu

Assistant Chief Statistician

#

NEW YORK UNIVERSITY
A private university in the public service

Yaw Nyarko
Director of Africa House
Professor of Economics
New York University
19 West 4th Street, Room 505
New York, NY, 10012

Telephone: +1 (212) 998-8928
Mobile (U.S.): +1 (917) 446-1679
Mobile (U.A.E.): +971-50-442-9525
Email: yaw.nyarko@nyu.edu
Web: www.yawnyarko.com
Africa House: www.nyuafricahouse.org

November 14, 2019

To Whom It May Concern:

If the proposal submitted by Dr. Van Holt entitled "Transforming tropical commodity supply chains for mitigating climate change and sustainable livelihoods" is selected for funding by NSF, it is my intent to collaborate and/or commit resources as detailed in the Project Description section of the proposal.

Sincerely,

Yaw Nyarko
Professor of Economics
Director, NYU Africa House
Director, NYU Center for Technology & Economic Development

RAINFOREST ALLIANCE

Tensie Whelan
Clinical Professor of Business and Society
Director, Center for Sustainable Business
NYU Stern School of Business, KMC 8-51
44 West 4th Street
New York, NY 10012

Dear Tensie,

If the proposal submitted by Dr. Van Holt entitled "Transforming tropical commodity supply chains for mitigating climate change and sustainable livelihoods" is selected for funding by NSF, it is my intent to collaborate and/or commit resources, and/or nominate a delegate from the Rainforest Alliance to participate on my behalf as detailed in the Project Description section of the proposal.

Warm Regards,

Han de Groot

Chief Executive Officer